GLACIER FEATURE DELINEATION AND PREDICATION IN RISHI GANGA REGION

A DISSERTATION

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I, Sarthak Chauhan, hereby declare that this dissertation titled "**GLACIER FEATURE DELINEATION AND PREDICTION IN RISHI GANGA REGION**" is the record of an original and independent study carried out by me under the guidance of Dr Ajanta Goswami in the Department of Earth Sciences, IIT Roorkee, submitted in partial fulfilment of the requirement for the award of the degree of Integrated Master of technology in Geophysical Technology.

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This is to certify the work displayed in this thesis titled "**GLACIER FEATURE DELINEATION AND PREDICTION IN RISHI GANGA REGION**" submitted by Mr. Sarthak Chauhan to Indian Institute of Technology, Roorkee in partial fulfilment of the requirements for the award of the degree is a record of original bona fide work of **INTEGRATED MASTER OF TECHNOLOGY** in **GEOPHYSICAL TECHNOLOGY** carried out by him with my guidance in the academic year 2021-2022.

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SARTHAK CHAUHAN

Abstract

Analysing the chronological glacier extent, the spatial pattern of the glacier changes and the use of machine learning to forewarn the future glacier coverage is crucial for understanding the cryosphere and the impact that might be caused on the system climate change. This study explains the glacier cover change and glacier retreat behaviour in the last nine years from 2013 to 2021 in the Rishi Ganga glacier region. Studies in this region became much more important after the region's glacier burst, showing changes in glacier retreat. The glacier cover was projected using the Normalize Difference Snow Index (NDSI) for the cloud-free Landsat data from 2013 to 2013. Furthermore, Artificial Neural Network-Cellular Automata have been used in this study to evaluate potential glacier cover. The findings of this investigation indicate that the glaciers on this mountain are receding. According to the data used in this research, the glacier's area has decreased by 56.1 percent between 2013 and 2021, with just Trisul and Nanda Devi Peak remaining. The glacier retreat on Rishi Ganga Basin was also found to be strongly associated with elevation, followed by aspect and slope. According to the forecasting simulation, the glacier will continue to melt (with a level of accuracy of 89 percent and Kappa value of 0.87). As a result, the Dunagiri glacier will be gone by 2025.

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Chapter 1. Introduction and Literature Review

According to D. H. Shugar (2021), around 30–50 percent of glaciers are snowfall feeders in the Himalayan region, supplying enough meltwater to rivers that originate from glaciated areas (D. H. Shugar et al., 2021). There is a large number of studies on glacier retreat, but most studies never go into detail about how glacier shrinkage happens when using a spatial analysis technique. Geographical factors may have an impact on glacier distribution, resulting in a variety of local features. This analysis, however, lacks a quantitative approach, and the results are not consistent. Simultaneously, understanding the change in mean climate and climate impact drivers requires assessing the regional response pattern of climate change at various global warming levels as reported in IPCC AR6, 2021.

There have been no studies that use machine learning techniques to estimate when the glaciers in Rishi Ganga may vanish accurately. This insight is also necessary for comprehending the cryosphere of the Rishi Ganga basin and the impacts of climate change on it. From 2013 to 2021, this study will use the Weight of Evidence to study and record the glacier change pattern in the Rishi Ganga basin. The Weight of Evidence was implemented to determine how input data and interpretive factors interact spatially (Fat et al., 2011). As interpretive information, this technique will investigate the relationship between glacier change and the morphological state of the Rishi Ganga basin in order to present the spatial factor for the process of learning in ANN-CA. The ANN-CA will then model the glacier change layer and its interpretive characteristics in order to predict future spatial coverage. This research will identify the research study of how the glacier retreat pattern on the Rishi Ganga basin has changed between 2013 and 2021, as well as glacier forecasting for each year from 2021 till it completely vanishes.

The research objectives of the study are:

[1] Analyse Spatiotemporal changes in glaciers using existing cryospheric algorithms.

[2] Estimate the spatial change patterns, on a pixel-by-pixel basis, for the Rishi Ganga basin using the cloud-removed Landsat dataset for 2003-2021.

[3] Prepare a glacier retreat model to perform trend analysis of the complete basin to predict annual glacier cover for upcoming years.

[4] In addition, also to create a GUI tool and high-performance python script, released as open-source, to aid researchers in replicating the preparation of such cloud-free datasets.

The first chapter presents the introduction, the research questions, and the literature review of earlier studies. A theoretical description of various components of the study is presented in the section. The second chapter describes the study area and dataset used for this study. This section details the Rishi Ganga basin situated in the state of Uttarakhand. The third chapter describes the methodology followed in this study. The fourth chapter discusses the results obtained at various steps of the study. The fifth chapter details about tools that were generated and published in the public domain during this study. The sixth and final chapter provides concluding remarks to the study.

1.1 Imagery Corrections

1.1.1 Geometric correction

Geometric correction is an essential technology in the processing and application of remote sensing images. The most common sensor models use collinear conditional equations to provide a rigorous image geometry model. However, the imaging geometry is quite complex because the attitude control of the satellite is limited during imaging. The exact orbit, attitude, sensor imaging parameters, and imaging method of the satellite must be understood for developing a model.

Stripe-like abnormalities in the along-track and across-track directions of the satellite path are a frequent phenomenon emerging from the architecture of current push broom satellites. Since such abnormalities have a distinct spatial frequency, it is possible to compensate by using pixels on stable ground. Another prominent cause of geometric errors is inaccuracies in the DEM used for orthorectification, which can result in apparent displacements over stable areas and errors in the measurements of true surface motion. The size of the

measurement error (e_d) is thus proportional to the incidence angles at which the two-satellite data were captured and the topographic model error.

$$e_d = e_h * tan\theta_1 - tan\theta_2$$

The above equation demonstrates how abnormalities can be decreased by selecting image pairs with low baselines and, as a result, small convergence angles. If a strong vertical component of the motion is to be expected, the DEM should ideally be updated for each date. While stereo-pairs or tri-stereo acquisitions from very high spatial resolution (VHSR) optical satellites are increasingly being used to update topographic information in Digital Image Correlation (DIC) processing chains (A. Stumpf, 2016), this is still not always possible.

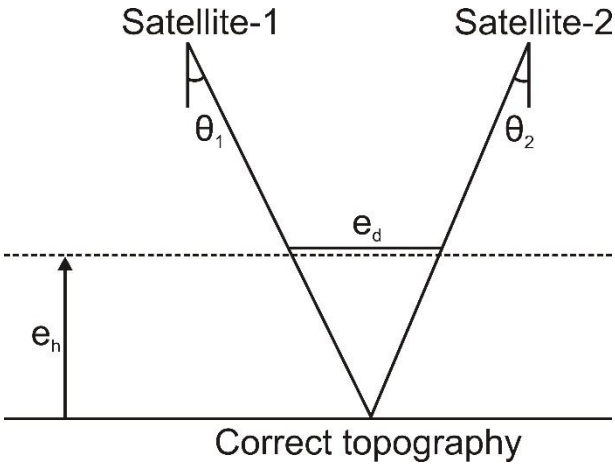

Figure 1.1 Relationship between incidence angles (θ_1, θ_2) at which two images (I_1, I_2) have been recorded, DEM error (e_h) and the resulting error in the measured displacement (e_d) for a stable point.

Random errors require more prolonged time series that may allow averaging their contribution out or detecting errors due to inconsistencies in the directional variance of the displacement field (A. Stumpf, 2016). Systematic errors in the DEM (like dependent on aspect and slope) can also be taken into account by fitting regression models over stable terrain, whereas random errors require a more prolonged time series that may allow averaging their contribution out or detecting errors due to inconsistencies.

Another postprocessing method is to replace outliers that are not compatible with a smooth motion field by filtering measurements with low confidence according to the employed distance/similarity measure or spatial filtering. For gravitational mass movements, in

particular, it may be hypothesized that the movement's direction will exactly approximate the gradient of the maximum slope, which can be used to eliminate vectors that deviate substantially from the slope's direction. Snow surfaces are often correlated with significant inaccuracies in DEMs obtained from photogrammetric approaches, making image matching difficult. Employing multispectral remote sensing imagery, snow masks can be generated to eliminate such areas and limit the number of false-positive matches.

1.2 Cellular Automata

Cellular automata are computational models that use computing to replicate simple rules; as a result of their interaction with entities as they evolve, these simple rules can form complex emergent phenomena. Von Neumann and Alam originally introduced the concept in the mid-twentieth century, and then a few decades later, the popular game of life brought interest to the subject beyond academia. Stephen Wolfram conducted a comprehensive investigation of cellular automata and published a book titled "A new kind of science" in the 1980s, proposing that cellular automata may permit a unique approach based on algorithm development. Modern science holds the belief that simple rules can only produce simple phenomena, and those complex phenomena must be the outcome of complex rules. The introduction of chaos theory in the last decade has revealed that this assumption is inaccurate, as simple systems like a pendulum have been shown to be able to construct complex and chaotic behaviour. Complexity may not be the result of complex rules but rather from the interaction of simple rules as they grow. Cellular automata are the scientific tools that embody this philosophy.

Cellular Automata does have a set of impediments that are required to be followed by any researcher:

1. There should be no explosive growth.
2. There should exist small initial patterns with chaotic, unpredictable outcomes.
3. There should be potential for von Neumann's universal constructures.
4. The rules should be as simple as possible whilst adhering to the above constraints.

The English mathematician Jhon Conway devised the Game of Life, which is played on a grid of square cells. It is possible for a cell to be living or dead. By leaving the square empty, a dead cell is indicated. Each cell has a neighbourhood that consists of all of its neighbouring cells, and the agent's behaviour is governed by four rules. A live cell with fewer than two live neighbours dies as if caused by under-population.

1. A live cell with two or three live neighbours lives on to the next generation.
2. A live cell with more than three live neighbours dies
3. A dead cell with exactly three live neighbours becomes a live cell, as if by reproduction.

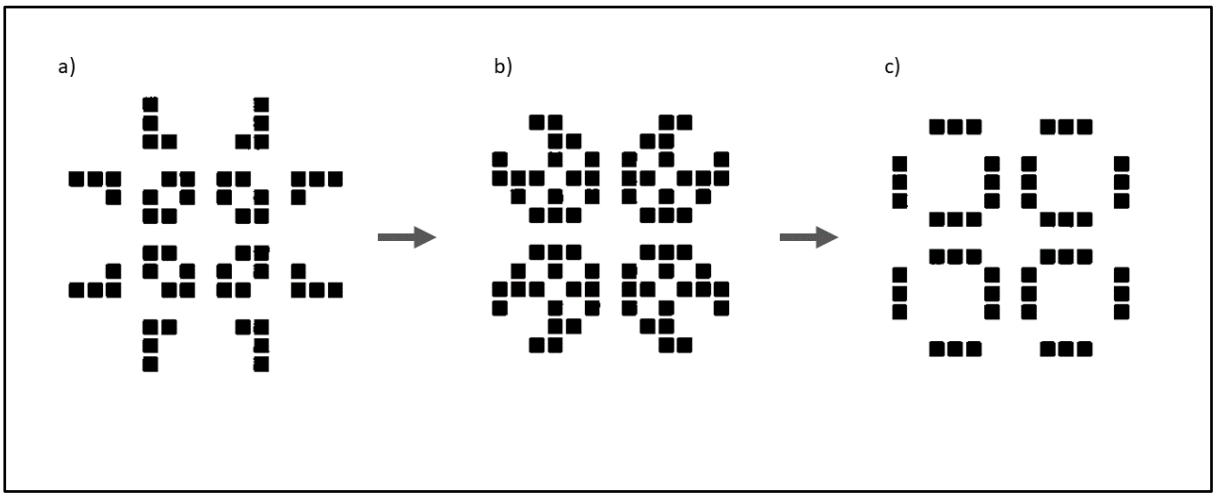

Figure 1.2 Game of Life, live-cell transition in accordance with the set rules.

The cellular order of classification is numbered from one to four in order of complexity that they can sustain as they go from stable to periodic oscillations to chaotic and complex patterns of behaviour. Class 1 patterns where all of the patterns evolve quickly to a stable and homogeneous state. The system develops into an oscillating structure, similar to a blinker, in Class 2 patterns. Almost all initial patterns evolve in a semi-random or chaotic structure in Class 3 patterns. The surrounding noise quickly destroys any stable structures that appear. Here, a group of cells gliding across the screen called Gliders are observed (an excellent example of emergence, as the simple rules producing them, can no longer be observed but instead an emergence structure gliding). Finally, automata may generate complex patterns that last for a long period while maintaining stable local structures. Cellular automata may model various real-world phenomena, including biological and chemical processes, using these complex structures. Since the advent of the game of life, new cellular automata have been

developed that can do all sorts of things, such as fractals. From the prospect of computation, the game of life can do anything that a computer can do.

The equations have been the dominant form of mathematical models throughout the history, through which so-called scientific laws of nature have been encoded. There are many applications for equation-based modelling, but they have their limitations. They present a static picture of the world in a state of permanent equilibrium; this is exhibited within economic models that describe the market as always moving towards an equilibrium state between supply and demand. But in reality, many complex systems like ecosystems and societies are only at or near equilibrium when they are dead. New things are constantly emerging and disrupting the equilibrium. Equilibrium of this kind is not well modelled by equations. It is best described by the evolutionary dynamics that shape them, with cellular automata again being well designed to capture this.

So it can be said that cellular automata are a computation modelling method based upon algorithms that iterate on simple rules to try and simulate complex phenomena. They are better suited to modelling phenomena that are the product of the evolutionary dynamics and have emerged out of the interaction between their parts, as is typically the case for complex systems.

1.3 Artificial Neural Network

An artificial neural network (ANN), commonly known as a neural network, is a computational model that can process information in order to solve problems, including classification and regression. Artificial Neural Networks were first established as highly simplified models of the human nervous system that could learn, generalise, and abstract. Artificial neural networks can not only acquire, preserve, and recognise patterns of information based on experience, but they can also retrain themselves for a dynamic environment, and they have better cognitive capabilities such as learning and memory (S. Chen and D. Tan, 2018).

As a result, they provide an interesting methodology with a wide range of possible applications, from earthquake investigations to portfolio management techniques (F. D. Pieleanu, 2016). ANNs are mathematical and analytical models that implement a simple pattern of interconnected parts to imitate the behaviour of biological neurons for the resolution of

complex and non-linear processes. The ability of ANNs to learn mathematical and statistical models through experience is its most prominent feature. This fundamental feature prevents ANNs from directly evaluating the mathematical relationship between inputs and outputs. ANN is an excellent prediction tool (Gaston H. Gonnet and Ralf Scholl, 2009.) robust tool in estimation (S. M. Mousavi, 2015.) to find the best combinations of input variables that affect output variables (D. K. Sharma and A. R. Rababaah, 2014). ANNs are thus a kind of black-box model in that it is impossible to know exactly how they achieve their output, and they are not programmed as a conventional calculation code; instead, they are trained using actual data through a learning process (Cinzia Buratti and Domenico Palladino, 2016).

1.3.1 Multi-input Neuron

A neuron may contain one or more input parameters ($x_0, x_1, x_2, \dots, x_n$) weighted by corresponding elements ($w_0, w_1, w_2, \dots, w_n$) of weight matrix W.

$$Net\ input\ (n) = w_0x_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n + \mathrm{b}$$

Or

$$n = \sum_{k=0}^{k=n} w_k x_k + b$$

Where b represents the bias or offset.

$$n = Wp + b$$

The output *n* (or net input) is passed into a transfer function (*f*), which yields the scalar neuron output a. The output for the neuron can be written as:

$$a = f\,(Wp + b)$$

Approximating the simple model to the biological neuron, the weight correlates to the strength of a synapse. The cell body is represented by the summation and transfer functions, while the signal on the axon is represented by the neuron output.

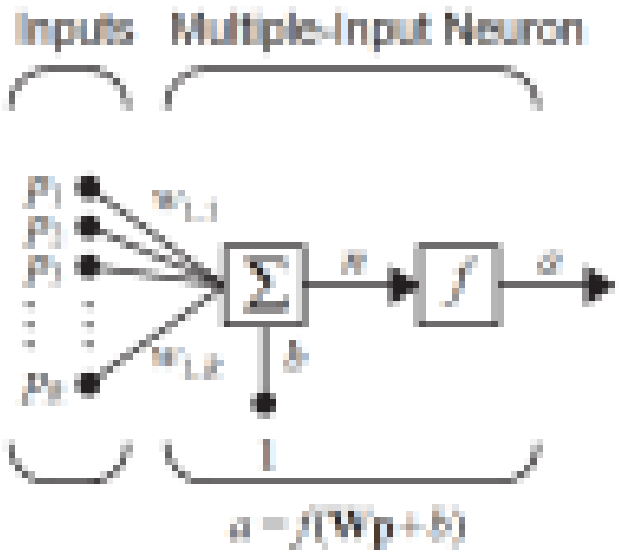

Figure 1.3 Multiple-Input Neuron

A neuron can do more than weight and sum the information it receives and transmit it to the next neuron with the help of the activation function. Such a process is simulated by the activation function. In actuality, neurons in the body do not always engage after receiving information; instead, they assess if the information has surpassed a threshold. The neuron is in a state of suppression if the value of the information does not outweigh this threshold. Otherwise, the neuron is activated, and the information is transmitted to the next cell.

A neuron in an ANN would reach the same conclusion as to whether and how to pass the information. Non-linear functions are usually utilized instead of thresholds as the activation functions of neurons to optimize their utilisation. Furthermore, by introducing non-linear activation functions, a neural network can theoretically meet any objective function, making it robust.

1.3.2 Linear Threshold Units

A linear threshold unit takes in scalar data as inputs ($x_0, x_1, x_2, \dots, x_n$), and finds the weighted sum of the input dataset.

$$SUM = w_0x_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n$$

or

$$SUM = \sum_{k=0}^{k=n} w_k x_k$$

A classification of the input data can be represented as "Hard Threshold":

$$f(SUM) = \begin{cases} -1, & SUM < 0 \\ 1, & SUM \geq 0 \end{cases}$$

The input x_0 is special; it is always -1. This effectively implements a non-zero threshold for the weighted sum of the actual inputs. At the boundary between the neuron outputting -1 and 1, SUM = 0, so

$$0 = w_0 x_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n$$

$$0 = -w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n$$

$$w_0 = w_0 x_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n$$

The special weight w_0 is often called the LTU's threshold. The plane of values $(x_1, x_2, x_3, \ldots, x_n)$ that leads to SUM equals zero is called the decision boundary because on one side the LTU outputs 1 and on the other side it outputs -1.

1.3.3 Transfer function

The transfer function can be a linear or a non-linear function of n. A particular transfer function is chosen to satisfy some specification of the problem that the neuron is attempting to solve. Various transfer functions are available, but the functions used in this study include:

The **linear transfer function** designates the output of the neuron as equal to its input.

The **hard limit transfer function** designates the output of a neuron to zero if the function assertion is less than zero, or one if its assertion is greater than or equal to 0. The hard limit transfer function can be used to generate neurons that classify inputs into two groups.

The **log-sigmoid transfer function** takes the input and suppresses the output of the neuron into the range 0 to 1. Because the log-sigmoid transfer function is differentiable, it is often utilised in multilayer networks trained with the backpropagation algorithm (H. Ullah and M.A. Kiber, 2018). Sigmoid function can be defined as:

$$a = \frac{1}{1 + e^{-n}}$$

1.3.4 Layers of Neurons

Previous research has observed that one neuron is not sufficient even with multiple input parameters; therefore, adding more neurons, like layers, operating in parallel increases efficiency. And each of these layers will have their weight matrix, bias vector, net-input vector and output vector. All of the neurons in a layer may not have the same transfer function; in ANN, a single layer of neurons with different transfer functions can be defined by combining two networks. Both networks would use the same inputs and generate some of the same outputs.

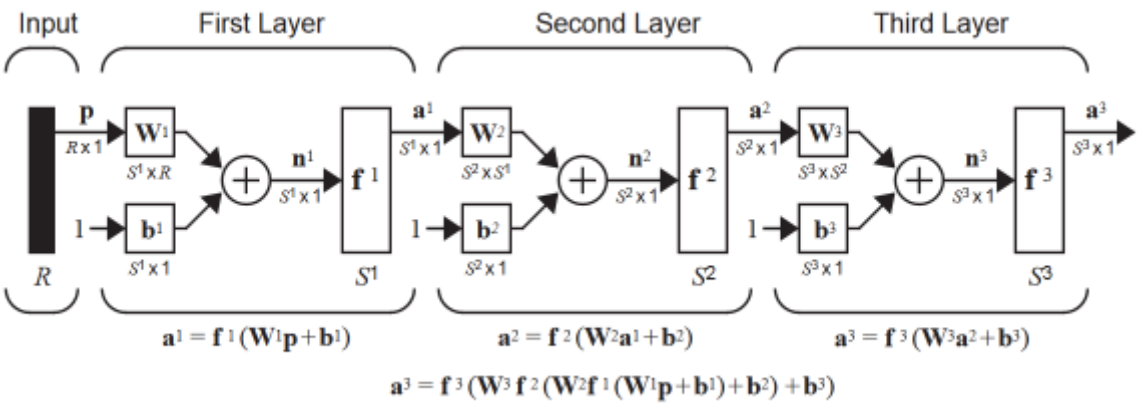

Figure 1.4 Multiple Layer Neuron

The output parameters for a layer are taken as the input value of the proceeding layer in the Neural Network model. The layer that provides the output for the whole network is named the output layer, while the other layers come under the hidden layer. Multilayer networks are more potent than single-layer networks as they are trained to approximate most functions arbitrarily well.

1.3.5 Learning Rules

The learning rule aims to train the network to perform some tasks, more specifically, modifying the weights and biases of a network. The neural network learning rules fall into three broad categories:

Supervised learning: It occurs when the model is given examples (the training set) of appropriate network behaviour. The network outputs are compared to the targets as the inputs are applied to the network. The learning rule is then used to adjust the network's weights and biases, bringing the network's outcomes relatively close to the targets. This supervised learning includes the perceptron learning rule.

Reinforcement learning: It is similar to supervised learning, except that instead of receiving the right output for each network input, the algorithm is only given a score. The score is a measurement of the network's performance over a set of inputs.

Unsupervised learning: During training, no target or output data is provided; weights and biases are modified in response to network input variables. By performing clustering processes, the algorithms learn to categorise the input patterns into a finite number of classes. The unsupervised learning approach includes vector quantisation.

1.3.6 Feature scaling

When the range of values is very distinct in each input parameter, we need to scale them to the standard level, making feature scaling an essential pre-processing step required for standardising/normalising the input data. Feature scaling of the dataset is performed to change the feature vector into a more suitable format for deep learning techniques (Thara D. K., 2019). In case when the values of one of the input parameters are very high compared to others, the impact of the input parameter with a higher value will be much higher as compared to the impact of other low valued inputs. Feature scaling can help reduce the time of execution as they directly influence the regression coefficients.

Also, the gradient descent can be achieved easily with the scaled dataset. The gradient descent uses one fixed learning rate for all $\theta's$, so a value must be selected based on the smallest

range of input values. Otherwise, the gradient descent might not converge for that small range. With that small learning rate, it takes ages for the large range to converge.

To keep the following in mind, scaling the inputs renders the error surface more spherical, whereas it would otherwise be an ellipse with very high curvature. Because gradient descent ignores curvature, having an error surface with a high curvature will cause one to take quite a lot of steps that are not always in the optimal direction. When the input parameters are scaled, the curvature is reduced, making methods that ignore curvature (like gradient descent) work much better. The gradient points right at the minimum when the error surface is circular (spherical), so learning is easy (Geoffrey Hinton).

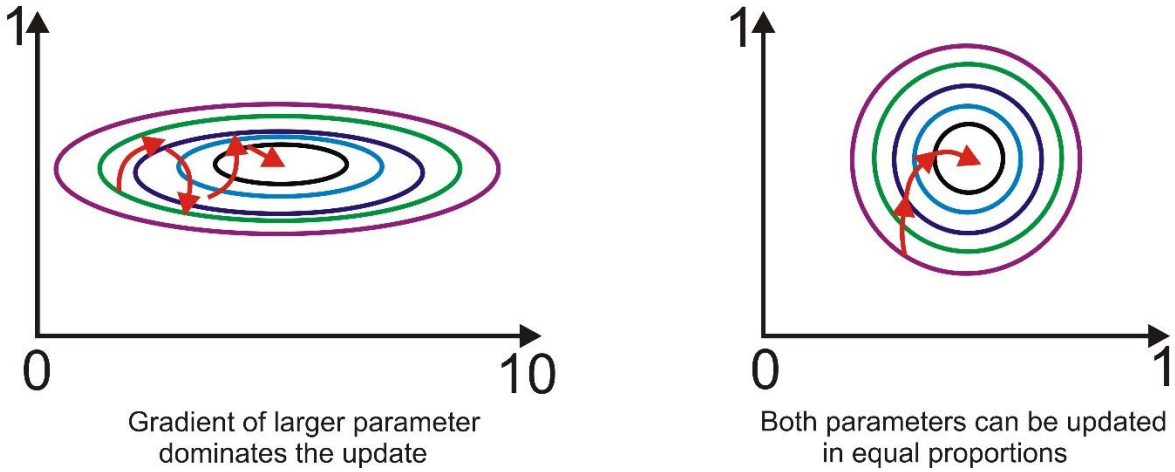

Figure 1.5 Effect of feature scaling on gradient.

1.3.7 Backpropagation Algorithm

The weights and biases of an artificial neural network are randomly initialised. The neural network is expected to have errors in delivering the right output due to random initialisation. It must limit the number of errors as much as feasible. To reduce these error values, one needs a system that compares the neural network's desired output with the output that incorporates errors and adjusts the weights and biases to get closer to the expected outcome after each iteration. Therefore, the network is trained to backpropagate and update the weights and biases.

As seen in Section 1.3.4, in a multi-layer network, the output of one layer becomes the input of the next layer.

$$a^{m+1} = f^{m+1}(W^{m+1}a^m + b^{m+1})$$

for m = 0, 1, 2, …, M-1.

where M refers to the number of layers in the network.

External inputs are collected by the neurons in the first layer, which provides the reference point for the equation above:

$$a^0 = p$$

The outputs of the neurons in the last layer are considered the network outputs:

$$a = a^m$$

Sensitivities s^m are required to be computed using the chain rule. This step gives the process its name backpropagation because it describes a recurrence relationship. The sensitivity at the layer m is computed from the sensitivity at the layer m+1, which can be expressed through ta Jacobian Matrix.

$$\frac{\partial n^{m+1}}{\partial n^m} = \begin{bmatrix} \frac{\partial n_1^{m+1}}{\partial n_1^m} & \frac{\partial n_1^{m+1}}{\partial n_2^m} & \cdots & \frac{\partial n_1^{m+1}}{\partial n_{s^m}^m} \\ \frac{\partial n_2^{m+1}}{\partial n_1^m} & \frac{\partial n_2^{m+1}}{\partial n_2^m} & \cdots & \frac{\partial n_2^{m+1}}{\partial n_{s^m}^m} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial n_{ns^{m+1}}^{m+1}}{\partial n_1^m} & \frac{\partial n_{s^{m+1}}^{m+1}}{\partial n_2^m} & \cdots & \frac{\partial n_{s^{m+1}}^{m+1}}{\partial n_{s^m}^m} \end{bmatrix}$$

$$\frac{\partial n_i^{m+1}}{\partial n_j^m} = \frac{\partial\left(\sum_{j=1}^{s^m} w_{ij}^{m+1} a_j^m + b_i^{m+1}\right)}{\partial n_j^m} = w_{ij}^{m+1}\frac{\partial a_j^m}{\partial n_j^m}$$

$$= w_{ij}^{m+1}\frac{\partial f(n_j^m)}{\partial n_j^m} = w_{ij}^{m+1} f'(n_j^m)$$

where,

$$[illegible]$$

Therefore the Jacobian matrix can be written as:

$$[illegible]$$

where,

$$\dot{\mathbf{F}}^{m}(\mathbf{n}^{m}) = \begin{bmatrix} \dot{f}^{m}(n_1^{m}) & 0 & \dots & 0 \\ 0 & \dot{f}^{m}(n_2^{m}) & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & \dot{f}^{m}(n_{S^m}^{m}) \end{bmatrix}.$$

The recurrence relation for the sensitivity using the chain rule can be expressed as:

$$\mathbf{s}^{m} = \frac{\partial \dot{F}}{\partial \mathbf{n}^{m}} = \left(\frac{\partial \mathbf{n}^{m+1}}{\partial \mathbf{n}^{m}}\right)^{T} \frac{\partial \dot{F}}{\partial \mathbf{n}^{m+1}} = \dot{\mathbf{F}}^{m}(\mathbf{n}^{m})(\mathbf{W}^{m+1})^{T} \frac{\partial \dot{F}}{\partial \mathbf{n}^{m+1}}$$

$$= \dot{\mathbf{F}}^{m}(\mathbf{n}^{m})(\mathbf{W}^{m+1})^{T}\mathbf{s}^{m+1}.$$

It is observed that the sensitivities are propagated backwards through the network from the last layer to the first layer. The backpropagation algorithm uses the approximate steepest descent technique. The only complication is that to compute the gradient, sensitivities must be backpropagated first. The starting point, s^m, can be derived from the last layer:

$$s_i^{M} = \frac{\partial \dot{F}}{\partial n_i^{M}} = \frac{\partial (\mathbf{t}-\mathbf{a})^{T}(\mathbf{t}-\mathbf{a})}{\partial n_i^{M}} = \frac{\partial \sum_{j=1}^{S^M} (t_j - a_j)^2}{\partial n_i^{M}} = -2(t_i - a_i)\frac{\partial a_i}{\partial n_i^{M}}.$$

Now, since:

$$\frac{\partial a_i}{\partial n_i^M} = \frac{\partial a_i^M}{\partial n_i^M} = \frac{\partial f^M(n_i^M)}{\partial n_i^M} = \dot{f}^M(n_i^M)$$

Therefore,

$$s_i^M = -2(t_i - a_i)\dot{f}^M(n_i^M).$$

This can be expressed in matrix form as

$$\mathbf{s}^M = -2\dot{\mathbf{F}}^M(\mathbf{n}^M)(\mathbf{t} - \mathbf{a}).$$

The error is not an explicit function of the weights in the hidden layers for the multilayer network. The derivatives are determined using the chain rule of calculus since the error is an indirect function of the weights in the hidden layers.

$$\frac{\partial E[e^2(k)]}{\partial w} = \frac{\partial E[e^2]}{\partial a} \cdot \frac{\partial a(n)}{\partial n}$$

For the approximate mean square error, the steepest descent algorithm is:

$$w_{i,j}^m(k+1) = w_{i,j}^m(k) - \alpha \frac{\partial \hat{F}}{\partial w_{i,j}^m},$$

$$b_i^m(k+1) = b_i^m(k) - \alpha \frac{\partial \hat{F}}{\partial b_i^m}.$$

Chain rule can be used in the above equations, as:

$$\frac{\partial \hat{F}}{\partial w_{i,j}^m} = \frac{\partial \hat{F}}{\partial n_i^m} \cdot \frac{\partial n_i^m}{\partial w_{i,j}^m}$$

$$\frac{\partial \hat{F}}{\partial b_i^m} = \frac{\partial \hat{F}}{\partial n_i^m} \cdot \frac{\partial n_i^m}{\partial b_i^m}.$$

The net input to layer *m* is an explicit function of the weights and bias in that layer:

$$z_i^n = \sum_{j=1}^{n-1} w_{ij}^n z_j^{n-1} + b_i^n.$$

Therefore

$$\frac{\partial z_i^n}{\partial w_{ij}^n} = z_j^{n-1}, \quad \frac{\partial z_i^n}{\partial b_i^n} = 1.$$

If we now define

$$\delta_i^n = \frac{\partial E}{\partial z_i^n}$$

Using equations obtained from the chain rule:

$$\frac{\partial E}{\partial w_{ij}^n} = \delta_i^n z_j^{n-1}$$

$$\frac{\partial E}{\partial b_i^n} = \delta_i^n.$$

We can now express the approximate steepest descent algorithm

$$w_{ij}^n(k+1) = w_{ij}^n(k) - \alpha \delta_i^n z_j^{n-1}$$

$$b_i^n(k+1) = b_i^n(k) - \alpha \delta_i^n.$$

In matrix form this becomes:

$$\mathbf{W}^n(k+1) = \mathbf{W}^n(k) - \alpha \delta^n (\mathbf{z}^{n-1})^T$$

$$\mathbf{b}^n(k+1) = \mathbf{b}^n(k) - \alpha \delta^n.$$

1.4 Markov Chain Algorithm

In modelling data for future prediction, the data requires a methodology to transition from one image to another; for situations like this, mathematician Andrey Markov came up with a mathematical system that hops from one set of values, called state, to another. The chain

model says that the behaviours of the model could form a state space, that is, a list of all the possible states. For this study, the possible states are whether there is a phase or not. In addition, the Markov chain model tells the probability of transitioning from one state to any other state.

This sequence of transitions meets the Markov Property, which asserts that the probability of transitioning to any given state is completely determined by the present state and the amount of time that has passed, not by the state sequence that came before it. Markov processes are memoryless due to their one-of-a-kind property. A random process with the Markov property is known as a Markov chain. A random process, also known as a stochastic property, is a mathematical entity made up of random variables. Discrete state-space (a collection of possible random variable values) or discrete index sets are both included in a Markov chain (example: time).

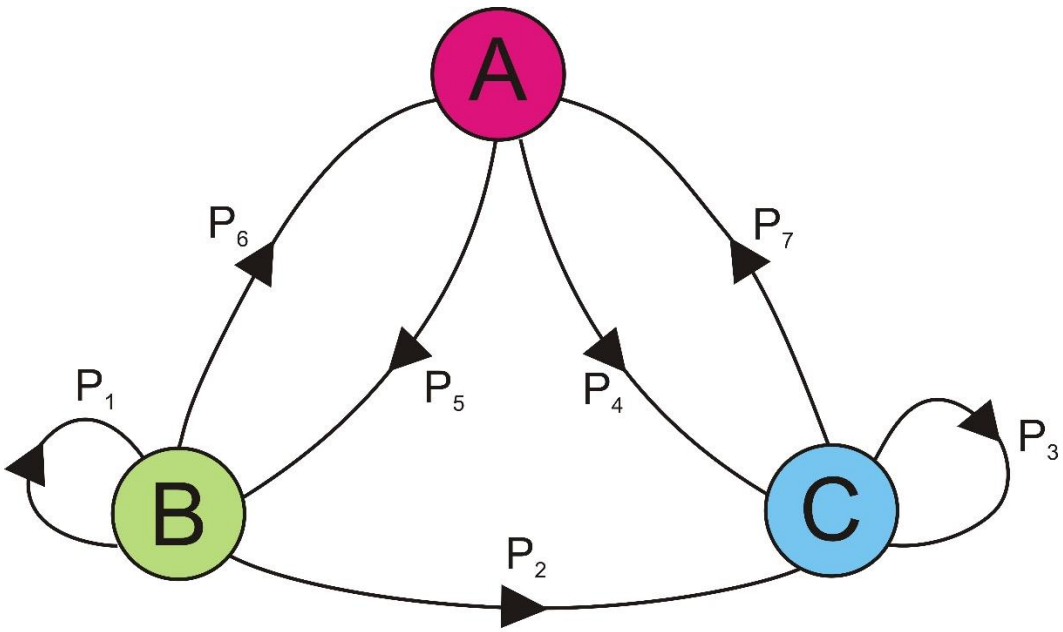

Figure 1.6 Markov graph of transition probabilities between states A, B and C.

There are some crucial properties of Markov chains are needed to be understood before going deep into modelling:

Reducibility: When a Markov chain can reach any state from any other state, it is irreducible. To put it another way, a Markov chain is irreducible if a series of steps with a positive probability exists between any two states.

Periodicity: In a Markov chain, a state is periodic if the chain may only return to it at multiples of an integer greater than 1. As a result, the chain can only return to state i at multiples of the period k, with k being the greatest such integer. State i is aperiodic if $k = 1$ and periodic if $k > 1$.

Transience and Recurrence: A state i is said to be transitory if there is a non-zero chance that it will never return to i, given that state i is taken as the starting point. If state i is not transient, it is recurring (or persistent). If a recurrent state is predicted to return in a finite number of steps, it is called positive recurrent; otherwise, it is called null recurrent.

Ergodicity: If a state i is aperiodic and positive recurring, it is said to be ergodic. An irreducible Markov chain is considered to be ergodic if all of its states are ergodic.

Absorbing state: If it is impossible to exit this state, it is called Absorbing. As a result, if $p_{ii} = 1$ and $p_{ij} = 0$ for $i \neq j$, the state i is absorbing. The Markov chain is an absorbing Markov chain if every state can reach an absorbing state.

1.4.1 Transition Matrix

When using matrix representation, the analysis of time-homogeneous Markov chains those whose transition probabilities are independent of time) becomes extremely effective. In particular, the formula given by the prior proposition becomes much more comprehensible. As a result, the following transition matrix completely represents the structure of a Markov chain:

$$T = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{bmatrix}$$

The nature of the elements that constitute transition probability matrices determines their properties. All Entries in transition matrix are non-negative, they must have a value between 0 and 1. The sum of all entries in each column of transition matrix is 1.

For some interval, say up to some period n, the transition matrix between any two successive periods, k and k-1, is the same as that for another pair, m and m -1. If looking for p for some k such that $1 < k < n$. Since the above is true, one can go from time period m-1 to time period m for any m.

$$T^{k-1}\overrightarrow{p_1} = \overrightarrow{p_k}$$

1.5 Weight of Evidence

The term "weight of evidence" usually directs to the interpretive methods of risk assessment or the risk statements that result from their use. Because of this concept's critical role in risk assessment, all stakeholders must understand its definition, applications, and implications (D. L. Weed, 2005). Science is at the heart of the Risk Assessment process, and science is made up of procedures: the research methods used to generate scientific data and the methodologies employed to summarise and evaluate that evidence. Anyone who is familiar with this technique will undoubtedly agree that its interpretive approaches may be improved. These methods are used to summarise and synthesise evidence from a variety of sources, including old and new studies and studies involving cellular systems. In addition to these apparent difficulties, adopting these methodologies necessitates the consideration of scientific and non-scientific values that are not usually stated explicitly. Even when the evidence is undisputed, assertions about a supposed hazard—for example, a chemical, drug, or consumer product—can disagree drastically.

Assuming a hypothesis or theory (H), and an event, evidence, or experimental outcome (E). A Bayesian assumes that probabilities like P(E|H&G), P(E|G), and so on, where G signifies background information, are meaningful. To save on notation, the background information is assumed and omitted from the notation, resulting in the various probabilities being indicated by P(E| H) and so on. These probabilities could be logical probabilities, also known as "credibilities," subjective or multi-subjective, and partially ordered, that is, interval-valued with upper and lower values or sharp (numerical) values. The discussion could be generalised to partially-ordered probabilities and weights of evidence but only with some loss of

transparency. If valid, any such generalisation should reduce to the sharp case when the intervals are zero width.

Let W(H:E) denote the weight of evidence favouring H provided by E, where the colon is read as "provided by". If some background information G is given all along, then the inscription can be extended to W(H:E|G). W(H:E) can be views as a function of P(E|H) and P(E|$\bar{H}$), such as f[P(E|H), P(E||$\bar{H}$)]. P(H), the starting probability of H, should be mathematically independent of the function f. The final probability of H, P(H|E), should only be determined by the weight of evidence and the starting probability.

$$P(H|E) = g[\ W(H:E), P(H)\ ]$$

$$P(H|E) = g[\ f[P(E|H), P(E|\ \bar{H})], P(H)\ ]$$

P(H) = x, P(E) = y, and P(H|E) = z we have the identity

$$z = g[f\left[\frac{zy}{x}, \frac{y(1-z)}{1-x}\right], x]$$

From this, it is deduce that f is a monotonic function of P(E|H)/P(E|$\bar{H}$) (Good, 1968, p. 141), and that it should, of course, be an increasing rather than a decreasing function. P(E|H)/P(E|$\bar{H}$) is a simple likelihood ratio if H and are simple statistical hypotheses, and E is one of the potential experimental outcomes appearing in the definitions of H and H. However, this is an uncommon case. In general, only a Bayesian considers this ratio to be significant. It's also known as a Bayesian likelihood ratio.

$$W\left[H:\left(EandE`\right)\right] = W(H:E) + W(H:E`|E)$$

Above equation can be approximated to:

$$W(H:E) = log\frac{P(E|H)}{P(E|\bar{H})}$$

It is seen that by using the applications of the product axiom of probability theory, namely P(A & B) = P(A).P(B|A), that

$$\frac{P(E|H)}{P(E|\bar{H})} = \frac{O(H|E)}{O(H)}$$

where 'O' denotes odds. The odds corresponding to a probability *p* are defined as *p*/(1-*p*).

Chapter 2. Study Area and Data Used

2.1 Snow-Cover Variability (Rishi Ganga)

Rishi Ganga is a river in the Chamoli district, Uttarakhand, India, with coordinates ranging from 79° 40' 7.5" E to 80° 2' 22.5" E and 30° 16' 14.5" N to 30° 32' 12.5" N. The South-Tibetan Detachment Fault (STDF) or Trans Himadari Fault (THF) surrounds the Rishi Ganga study basin in the north-eastern direction, passing proximally to the Nanda Devi peak, while the Vaikrita Thrust (VT) flows through Reni settlement in the southwest (Valdiya, 1980; Spencer et al., 2012a; Jain et al., 2014; A. K. Taloor et al., 2021).

The Rishi Ganga River, which has a drainage basin of roughly 625 square-kilometre, is nourished by the Uttari Nanda Devi glaciers, which are located at the base of the Nanda Devi Biosphere Reserve (NDBR). The river flows in a glacially carved valley until it reaches 2500 metres, then through a heavily engraved fluvial valley until it reaches the Dhauli Ganga River near Reni settlement. Three large glaciers cover 240 square-kilometre (50 percent of the Rishi Ganga basin) and have high monthly average insulation, surface temperature, and atmospheric temperature in the Rishi Ganga basin. The Raunthi River has a 90 square-kilometre drainage basin, with glaciers covering approximately half of the upper part of the drainage basin. At the height of roughly 2300 metres above sea level, the Raunthi River rises from the snouts of the Raunthi and Nandi Peak glaciers and enters the Rishi Ganga. A number of massive glaciers drain the slopes of Nanda Devi, Trisul, and other high Himalayan peaks at the headwaters of the Rishi Ganga Basin in India. Kumar et al. (2020) investigated the response of glaciers in the Rishi Ganga basin to climate change. The climate in the Rishi Ganga Basin is temperate, with considerable snowfall in the winter. The temperate, alpine, sub-alpine, moraine environments and many melting glaciers are all part of the Greater Himalayas.

A catastrophic debris-laden flood originating from the Raunthi River wreaked havoc on two hydroelectric plants in Dhauli Ganga and Rishi Ganga on February 7, 2021, extinguishing over hundreds of people (N. Rana et al., 2021). According to preliminary observations, the ice/rock wall break occurred along the Nandi Peak glacier's north-facing slope, resulting in the scavenging of roughly $7.9 \times 10\ 7\ m^3$ of sediments mixed with the

avalanche ice. The observed level was 1388.65 m, which is higher than the danger (1383 m) and warning levels (1382 m).

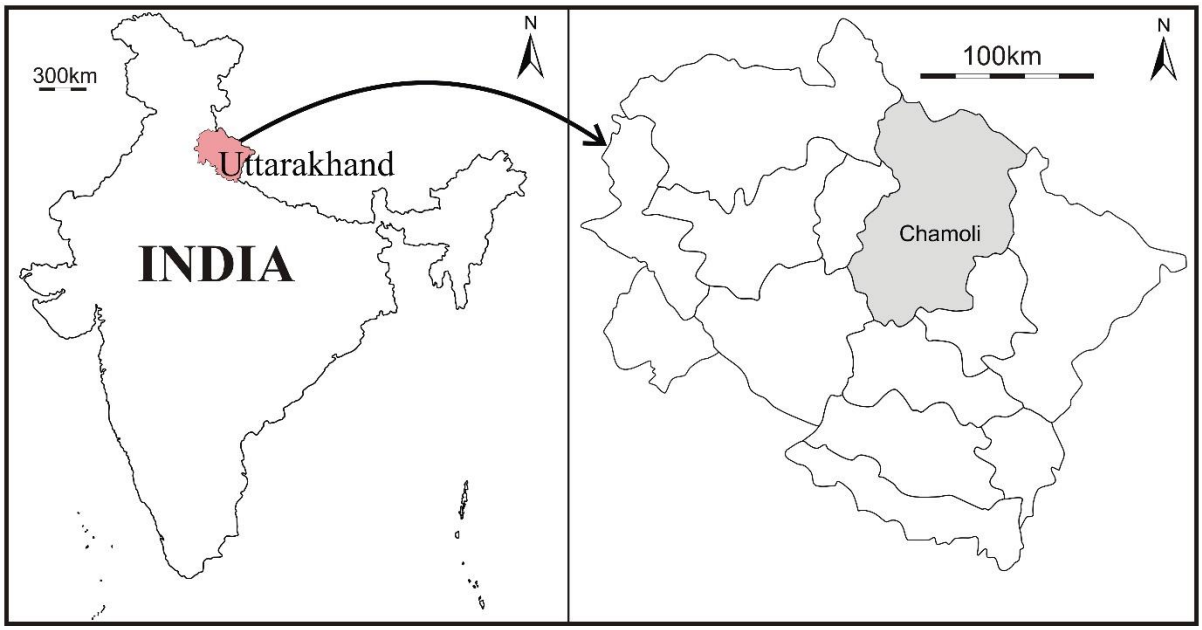

Figure 2.1 The boundary map of Uttarakhand shows the position of the Chamoli District.

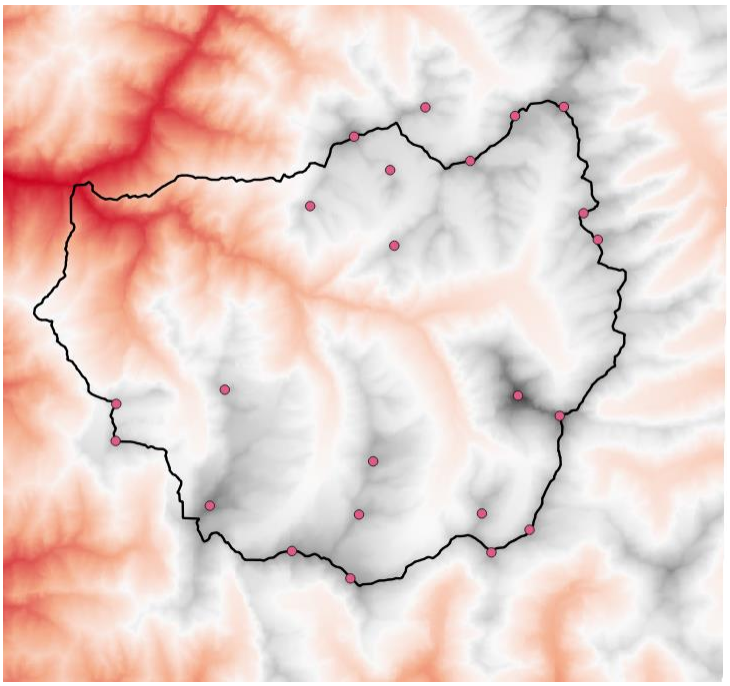

Figure 2.2 The Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) shaded terrain model-based elevation map of the Rishi Ganga Basin. The points on the map shows the major mountain peaks in the region.

2.2 Datasets

2.2.1 Multispectral Imagery

Since the glacier area has a high albedo, high reflectivity in the visible wavelength range of 0.4–0.7 m, medium reflectivity in the near-infrared range of 0.8–2.5 m, and low reflectivity in the thermal-infrared range of 2.5–14 m (Rees, 2003; Veettil & Wang, 2018), multispectral imaging is the best method for satellite glacier delineation (Kargel et al., 2014). This study used cloud-free images from the Landsat 7 Enhanced Thematic Mapper (ETM), and Landsat 8 Operational Land Imager to approximate the temporal glacier cover (1980-2021) in the Rishi Ganga region. However, the collected cloud-free images were not all found to be as effective due to image availability. This image collection could cause the classification of images to have a seasonal impact.

Table 1. Summary of Satellite Imagery Datasets utilized in this study.

Multispectral Dataset	Dataset Criteria	Acquisition Date	Spatial Resolution	Resources
Landsat 8 Operational Land Imager (OLI)	Glacier cover 2013	29 October 2013	30m	USGS Earth Explorer (http://earthexplorer.usgs.gov/)
	Glacier cover 2014	14 September 2014		
	Glacier cover 2015	03 October 2015		
	Glacier cover 2016	19 September 2016		
	Glacier cover 2017	24 October 2017		
	Glacier cover 2018	27 October 2018		
	Glacier cover 2019	14 October 2019		
	Glacier cover 2020	16 October 2020		
	Glacier cover 2021	03 October 2021		

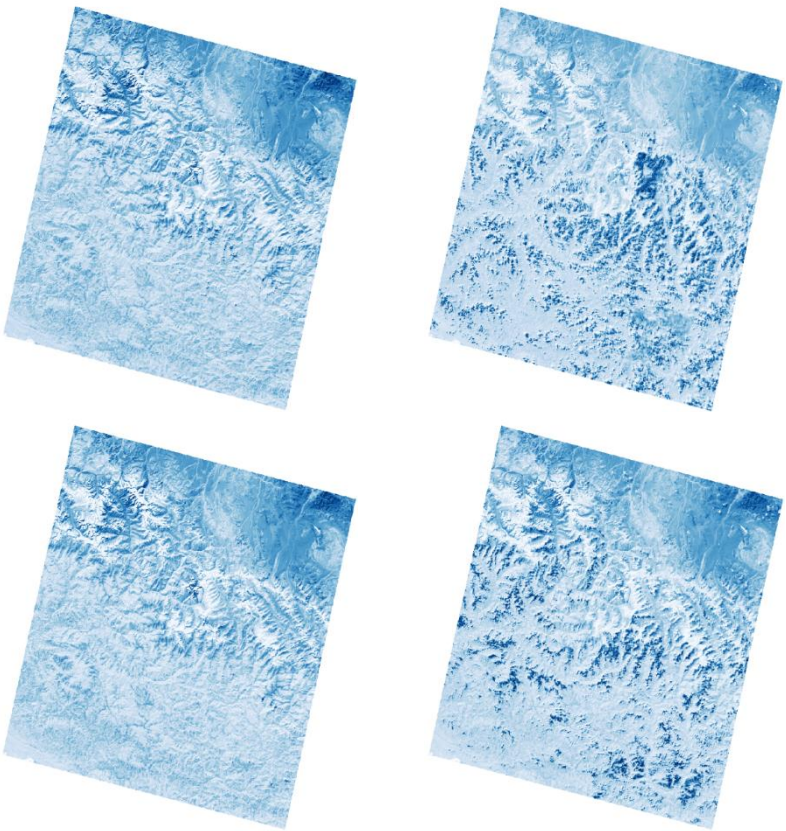

Figure 2.3 Landsat imagery for lower Himalayan Region for different year: a) 2013, b) 2014, c) 2020 and d) 2021.

2.2.2 ASTER Global DEM (ASTER GDEM)

Yang et al. (2015) and Yalcin (2020) proposed that topographic conditions, like climatic factors, are essential factors in glacier retreat spatial variation. Furthermore, improved climate change information requires an understanding of the spatial pattern response to glacier retreat (IPCC AR6, 2021). The extraction of Digital Elevation Model (DEM) data will deliver information about the morphological conditions of Rishi Ganga region as an explicatory criterion to analyse the glacier change pattern.

The ASTER (Advanced Spaceborne Thermal Emission and Reflection Radiometer) instrument lies onboard the Terra spacecraft. It can capture along-track stereo data using its pair of telescopes, one of which is nadir-viewing, while the other is backwards-viewing. The ASTER Global Digital Elevation Model (ASTER GDEM) is a near-global Digital Elevation Model (NASA/METI/AIST/Japan Space systems & US/Japan ASTER Science Team, 2019). The ASTER GDEM is a cloud corrected DEM product due to various corrections that

were used during the generation of the dataset. Initial corrections include cloud masking, statistical corrections, and referencing existing DEMs for correction and replacement of anomalies or missing data. The ASTER GDEM version 3 (ASTER GDEM V003) employs approximately 1.88 million input scenes to generate the final DEM output.

A GDEM error mask is also generated as part of the Quality Assessment process. Various voids (erroneous errors) are created when the error mask is generated. To fill these voids, an alternate GDEM (preferably an earlier ASTER GDEM) is used. The geolocation error is around 0.3 metres (west) and 5.4 metres (east) (North). The elevation error standard deviation is 12.1 m when compared to the GDEM V2 (NASA/METI/AIST/Japan Space systems & U.S./Japan ASTER Science Team, 2019).

Table 2. Summary of Digital Elevation Model and shapefile dataset utilized in this study.

Dataset	Morphological Criteria	Spatial Resolution	Resources
Digital Elevation Model (DEM)	Aspect	15 m	USGS Earth Explorer (http://earthexplorer.usgs.gov/)
	Elevation		
	Slope		
Basin Shapefile	-	-	RGI 6.0

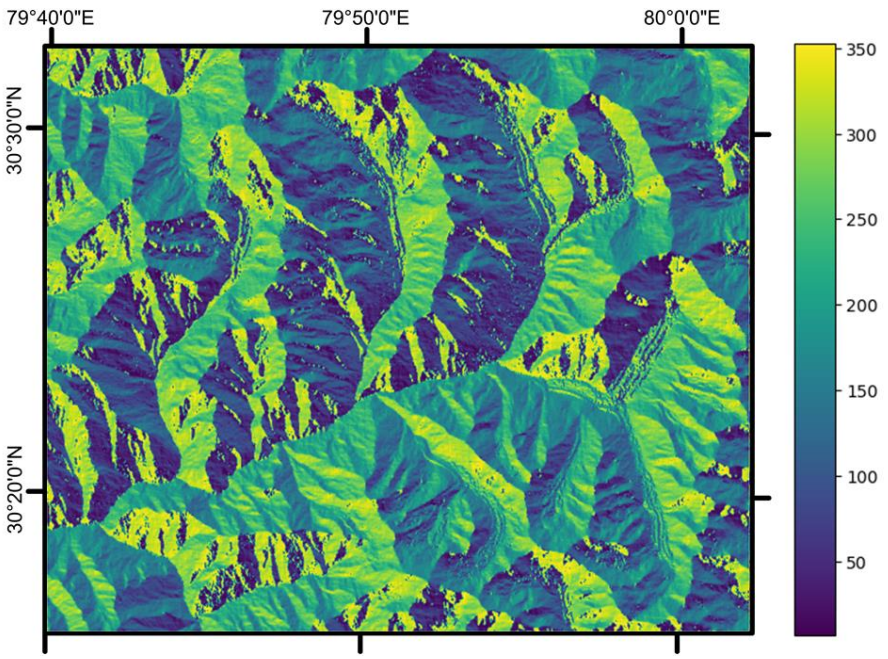

Figure 2.4 Aspect map (in degrees) for Rishi Ganga Basin.

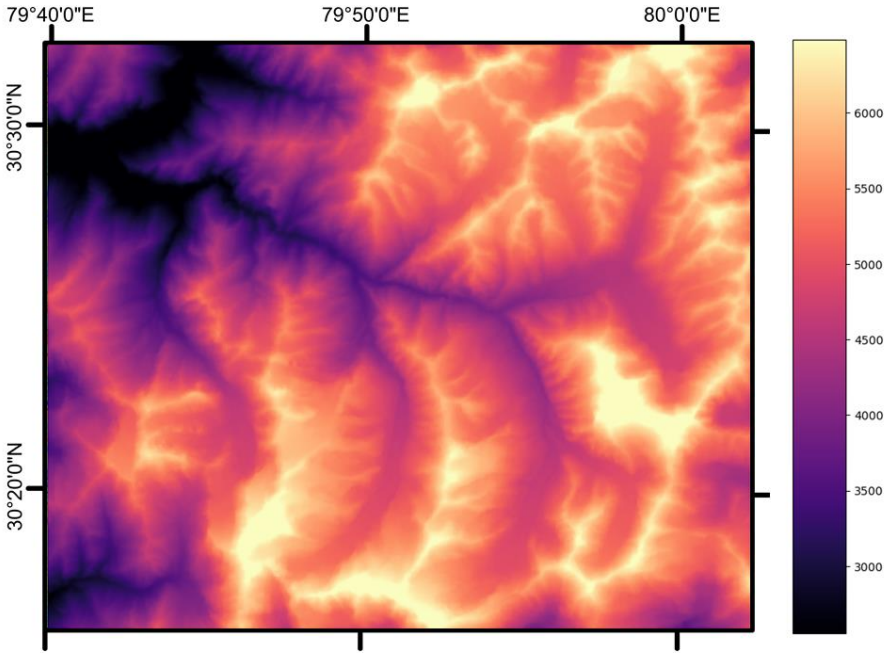

Figure 2.5 Elevation map (in meters) for Rishi Ganga Basin.

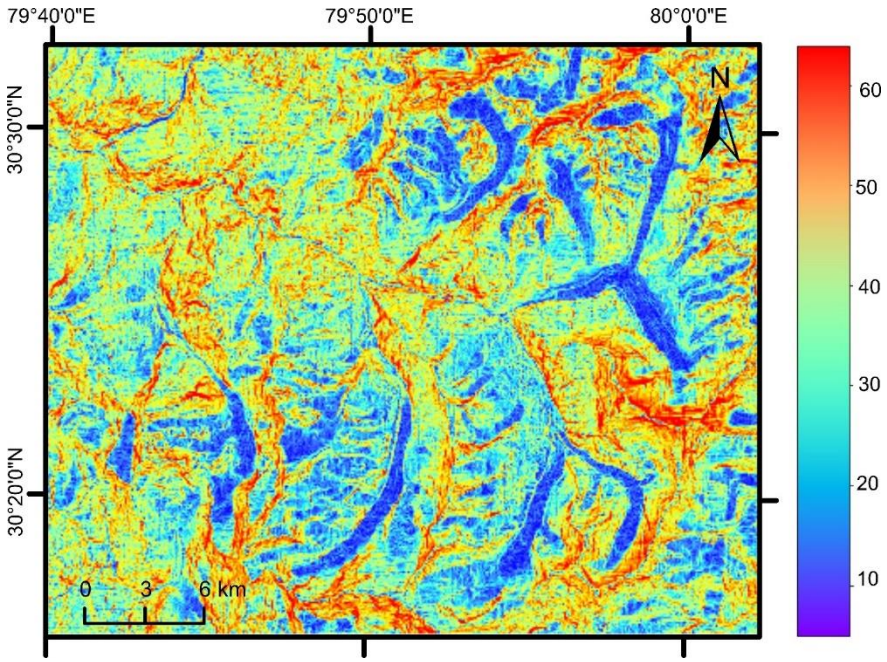

Figure 2.6 Slope map (in degrees) for Rishi Ganga Basin.

Chapter 3. Methodology

3.1 Feature Scaling

Veettil & Wang (2018) proposed that the glacier retreat can be influenced by morphological characteristics, such as the slope and aspect of that particular area. Another study claims the influence of slope, aspect, and elevation on glacier shrinkage in the mountain region (Hock (2005), Jiskoot et al. (2009). Another study by Muhamad I. J. Putra (2021) added distance from the mountain peak as an additional feature, but satisfactory results were not observed for a prolonged period of time. Given these findings, aspect, elevation and slope are used as the features for this study extracted from the DEM data. Dilts (2009) suggested that all these defining features are conditionally independent of the dependent variables.

As the study deals with extracting glacier retreat using these features, feature scaling becomes a fundamental step in order to get the expected results.

Table 3. Demonstrative morphological dataset processed through feature scaling.

Morphological Parameter	Number of Groups	Classification criteria
Aspect	8	North, Northeast, East, Southeast, South, Southwest, West, Northwest
Elevation	10	Less than 3000, 3000-3480, 3480-3960, 3960-4440, 4440-4920, 4920-5400, 5400-5880, 5880-6360, 6360-6840, greater than 6840
Slope	5	0-2, 3-10, 11-30, 31-40, greater than 40

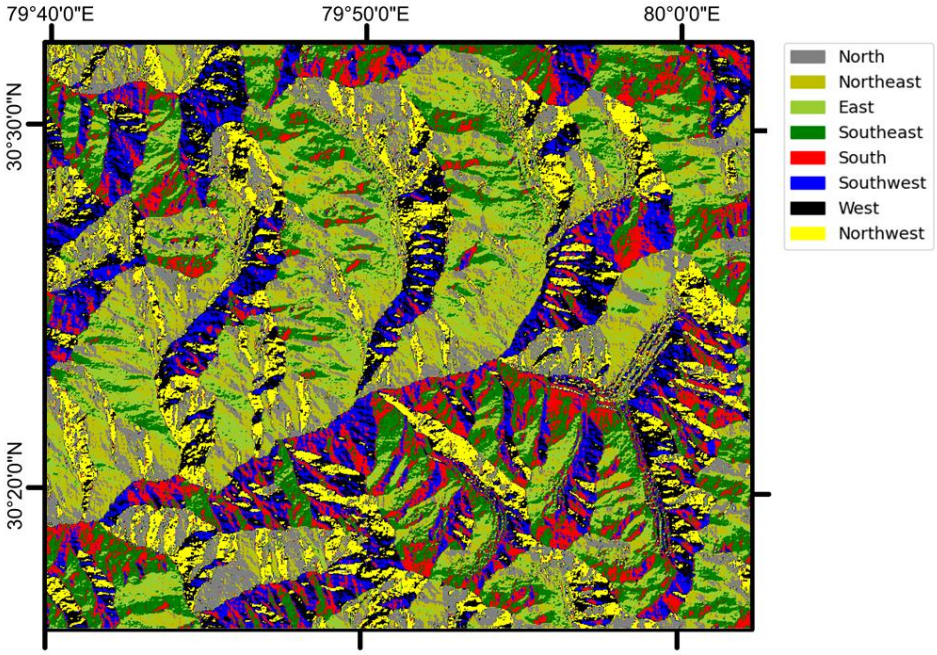

Figure 3.1 Aspect evaluated using the interpretivist approach used through feature scaling.

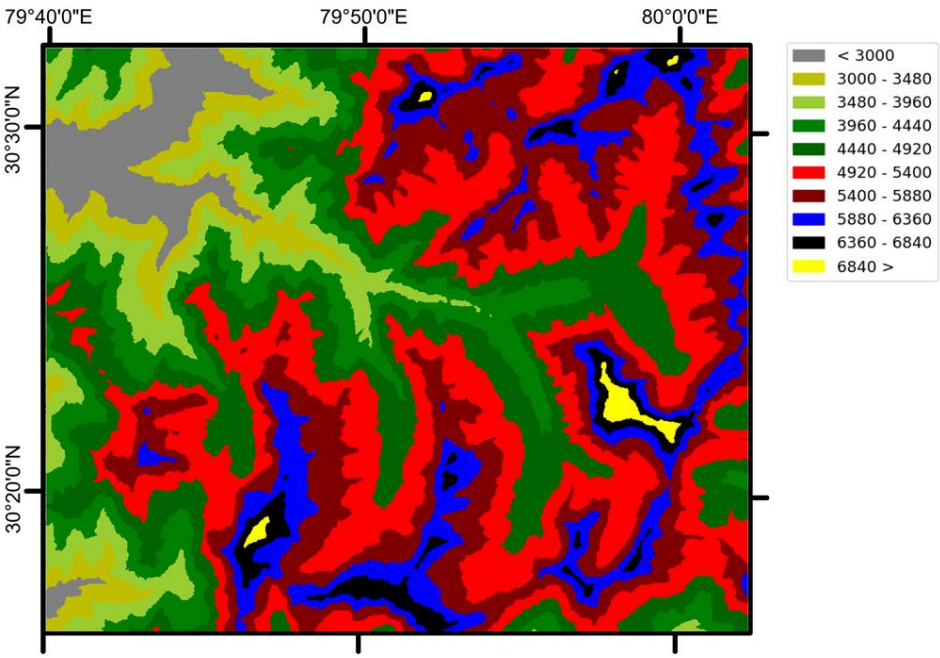

Figure 3.2 Elevation evaluated using the interpretivist approach used through feature scaling.

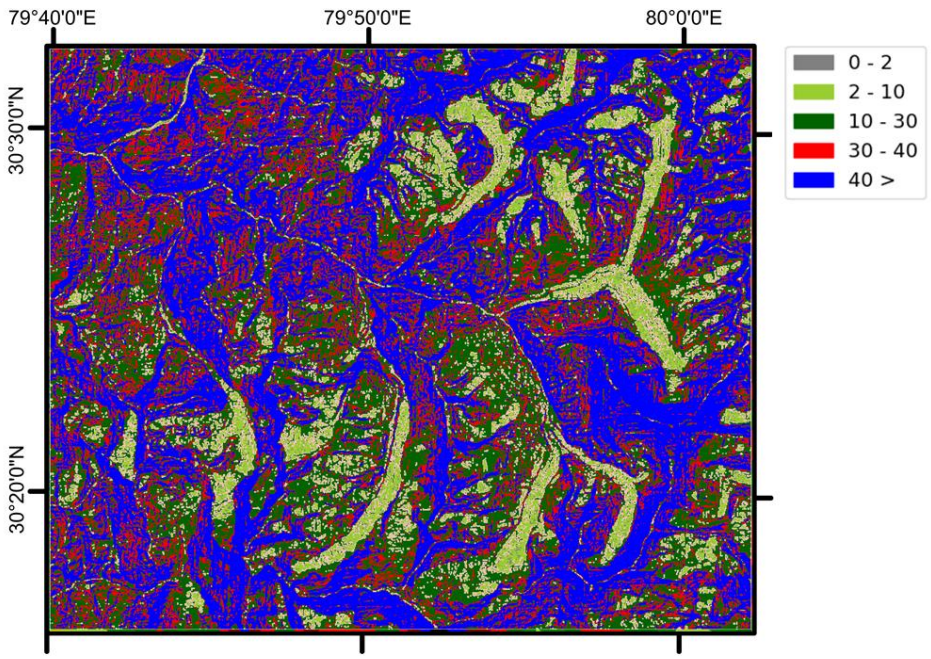

Figure 3.3 Slope evaluated using the interpretivist approach used through feature scaling.

3.2 Image Processing

The multispectral satellite image dataset required as an input for modelling has been extracted from the Landsat 7 EMT and Landsat 8 OLI series, and the images are required to be corrected before glacier mapping. The pre-processing step of Landsat data may require geometric correction, radiometric and atmospheric correction. While constructing a model, the exact orbit, attitude, sensor imaging parameters, and imaging technique must be known, but the satellites have limited attitude control resulting in a very complex imaging geometry. Because satellite imagery is prone to geometrical distortions, it cannot be used for precise terrain mapping, particularly in Spatio-temporal analysis. Various factors, such as refraction, ground curvature, terrain topographical effect, or terrain correction, might create such errors (Richter, 1998). The main goal of performing radiometric corrections is to eliminate irregularities in image brightness levels, which might make it difficult to understand, process, and analyse digital remotely sensed images statistically. Atmospheric adjustment (Liang et al., 2001; Chavez, 1996), sun elevation correction, and earth-sun distance correction (Richter, 1997) are all examples of radiometric correction, as are simpler techniques such radiometric normalisation (Canty et al., 2004) and histogram equalisation (Demirel et al., 2009).

Top and Bottom atmospheric correction of all the bands of Landsat images is required to avoid the path radiance and band noise (Kargel et al., 2014) and obtain the surface reflectivity value. The following equation is used to convert the digital number (DN) value to atmospheric radiance in this technique:

$$L\lambda = \frac{SR_k x\ DN}{DN_{max}}$$

Most of these corrections are already counted for in satellite imageries from Landsat 8 OLI or newer, but it becomes necessary to take care of data from satellites older than that of Landsat 7.

In contrast to previous studies that used a digitising technique (Klein & Kincaid, 2006; Kincaid, 2007) and image enhancement to extract glacier area, this study uses image processing techniques and a supervised classification approach to extract glacier area (Muhamad Iqbal Januadi Putra, 2021; Dorothy K. Hall (Veettil & Wang, 2018; Veetiil & Kamp, 2019). Image enhancement and modification, according to Rees et al. (2003), are required to maximise the separation of a glacier surface from a non-glacier object. The reflectance of snow and ice in the visible (VIS), near-infrared (NIR), and shortwave infrared (SWIR) areas of the spectrum has been reported to differ significantly (Kargel et al., 2014); therefore, the glacier area extraction is done using the band ratio Normalise Difference Snow Index (Rees, 2003; Veettil & Wang, 2019). As a result, NDSI will give the glacier surface a high reflectivity (Rees et al., 2003).

$$NDSI = \frac{Visible\ Green - SWIR}{Visible\ Green + SWIR}$$

Any pixel with an NDSI value greater than 0.0 is considered to have at least some snow. However, relying entirely on NDSI to categorise snow has a number of drawbacks, including the misclassification of salt pans and cloud edges as snow, as well as the labelling of factual

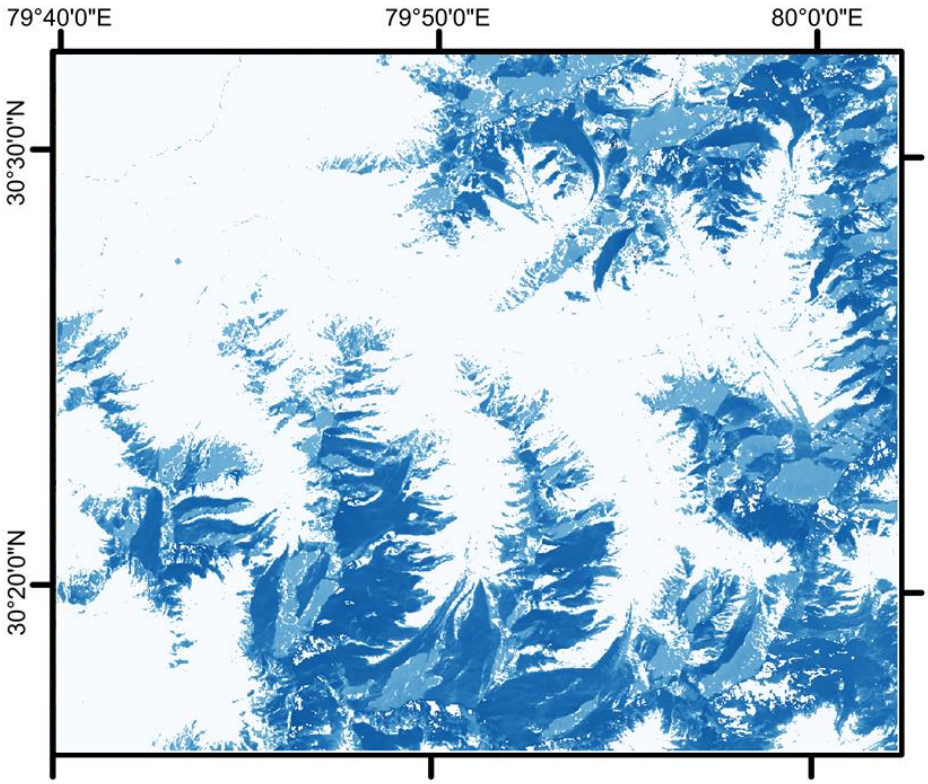

Figure 3.4 Snow cover map for Rishi Ganga region.

snow pixels as non-snow due to poor cloud conditions. Dorothy K. Hall et al., 2002 apply several screens and filters to the NDSI-based snow classification data to solve this challenge. If any of these displays are true for a classified snow pixel, that pixel is reclassified as non-snow.

The screens that are applied are as follows:

1. Low visible reflectance screen – It is based on the assumption that the reflectance must be greater than a determined minimum reflectance value for the NDSI algorithm to appropriately identify snow. If a pixel has reflectance that is less than this number, it is categorised as no-snow. If band two reflectance is less than 0.10 or band four reflectance is less than 0.11, the pixel is regarded to have low visible reflectance, and its value is inverted.
2. Low NDSI screen – If the NDSI value is between 0.0 and 0.1, the pixel is also inverted to no-snow.

3. Estimated surface temperature and surface height screen – if a pixel is detected at the height of <1300m and the temperature (MODIS band 31) is $\geq 281\ K$. Suppose the temperature of a $\geq 1300m$ pixel is detected to be $\geq 281\ K$, while the pixel value is not inverted. In that case, the pixel is marked as "Unusually high temperature" in the Quality Assessment (QA) flags that are imported along with the dataset.
4. High SWIR reflectance screen – Snow has characteristically low SWIR reflectance. If a snow pixel has a SWIR reflectance between 0.25-0.45, while it is not reversed as non-snow, it is marked as "unusually high SWIR" in the QA flags dataset. If a snow pixel has a SWIR reflectance value of >0.45, that pixel is inverted as non-snow.
5. Solar Zenith Screen – In low illumination conditions, the performance of NDSI for the detection of snow decreases drastically. Hence, in the QA flags dataset, low illumination flags are set to true if the solar zenith is > 70°. If the solar zenith is ≥ 85°, those pixels are marked as Night.

The binary classification will produce two types of information as a result of the categorization. The glacier area (positive/1) and non-glacier area (negative/0) were the two binary classes.

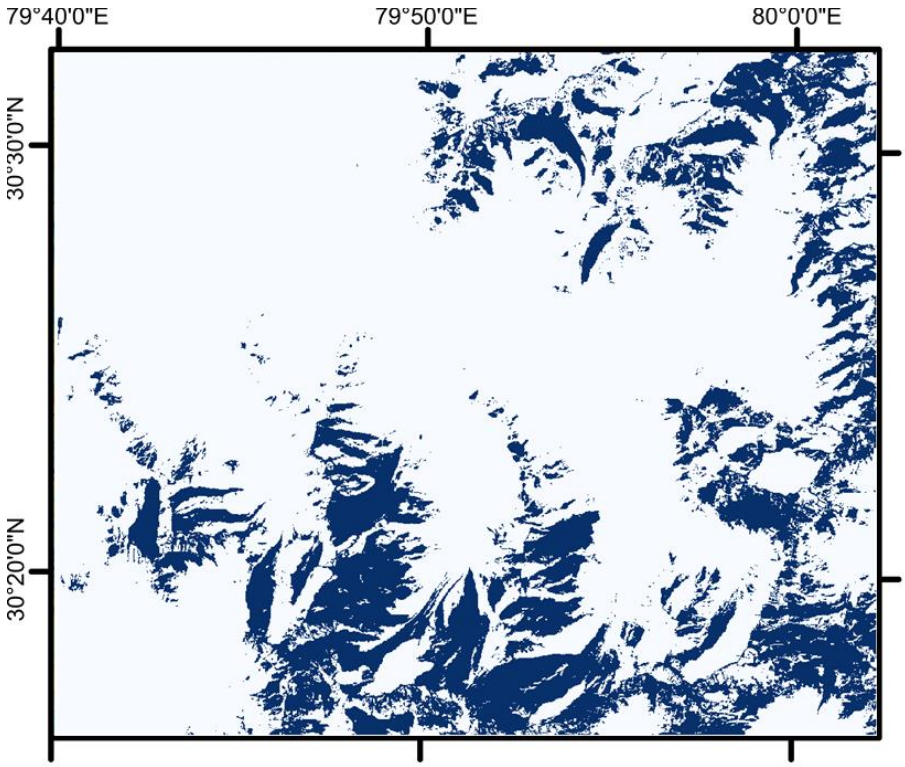

Figure 3.5 NDVI Classification of the study area.

3.3 Weight of Evidence

The Weight of Evidence (WoE) is a machine learning tool for determining the degree (probability) of the spatial correlation between predictive factors and dependent variables (in this case, glacier change) (Bousta & Brahim, 2018). During the feature engineering phase, the spatial components for this model were determined The application of WoE may be accomplished primarily by utilising the cross-tabulation technique through Geographic Information System (GIS) analysis, particularly for the binary classification feature. The cross-tabulation of binary variable evidence and its class spatial element is introduced by Matthew et al. (2007).

$$W^{+} = log_{e}\frac{P(X|D)}{P(X|\bar{D})}$$

$$W^{-} = log_{e}\frac{P(\bar{X}|D)}{P(\bar{X}|\bar{D})}$$

$$C = W^{+} - W^{-}$$

The degree of association between the glacier retreat pattern from 2013 to 2021 and the elucidative evidence, namely the morphological characteristics of Rishi Ganga Basin, are calculated using WoE in this work. The WoE algorithm is run in unit cells to determine the spatial footprint of an occurrence (Dilts et al., 2009). The outcomes of this procedure will yield a probability value for the degree of relationship (C, W+, and W-) between the glacier cover change and the predictive factors. In WoE, the C value can be anywhere between -∞ and +∞ (-∞ < x < +∞). The amount of association between the binary evidence and the binary explanatory class is represented by the C value. Meanwhile, the + and – signs in the C value denote the association's direction (positive or negative). The higher the C value, the more significant the association between glacier retreat and the characteristics analysed (Asmael, 2015).

One of the most effective methods for choosing relevant variables in a predictive model is to employ Information Value (IV). It aids in determining the relative importance of factors. It is a metric for separating binary outcomes, including whether or not glacier data is accessible.

$$IV = \sum (P(X|D) - P(\bar{X}|\bar{D})) * C$$

Table 4. Information Value range and its importance.

Information Value	Predictive Strength
less than 0.02	Unsuitable for forecasting
0.02 to 0.1	Unreliable predictor
0.1 to 0.3	Medium predictor
0.3 to 0.5	Effective predictor
greater than 0.5	Most Effective predictor

The weights of evidence analysis can be carried out using the Arc SDM spatial data modeller extension for ArcGIS (Sawatzky et al. 2004) and the modelling technique provided by Raines, Bonham-Carter, and Kemp (2000). To begin, a digital spatial repository including all of the evidence layers, as well as the research area and training points, was developed. The relationship between the training set and the evidence, including which layers of evidence are predictive. The natural breaks method (Sawatzky et al. 2004) can delimit classes, calculate weights, and classify data into positive as well as negative prediction classes. The evidence layers were then combined to provide posterior probability and confidence maps.

The employment of conditionally dependent evidence layers can lead to exaggerated posterior probability estimates (Agterberg and Cheng, 2002). However, this assumption is likely violated in nearly every weight of evidence analysis which was onbserved by Bonham-Carter 1994. The conditional dependence test based on a one-tailed test of the null hypothesis that conditional dependence is equal to zero (Agterberg and Cheng, 2002). A score of more than 50 percent indicates that conditional dependency is likely, and a value of more than 95 percent indicates that the null hypothesis (conditional dependence) should be rejected.

3.4 Glacier Retreat Prediction Algorithm

An artificial Neural Network (ANN) is a machine learning model for spatial predictive mapping in which the correlation between dependant and predictive variables is complicated (Noack et al., 2014; Bui et al., 2015). This study used Artificial Neural Network to forecast the spatial glacier patterns in the future, where the Landsat data and the morphological dataset is used in the input layer while the Glacier retreat maps act as the output layer.

For simulating glacier change, an ANN will analyse the transition probability of glacier cover using neurons in the input layer and a hidden layer associated with a particular weight. Cellular Automata, on the other hand, analyses the transitional matrix information from ANN simulations to process future glacier coverings (Saputra & Lee, 2019). ANN-CA has been widely employed in geospatial domains, particularly in the land use/land cover sectors.

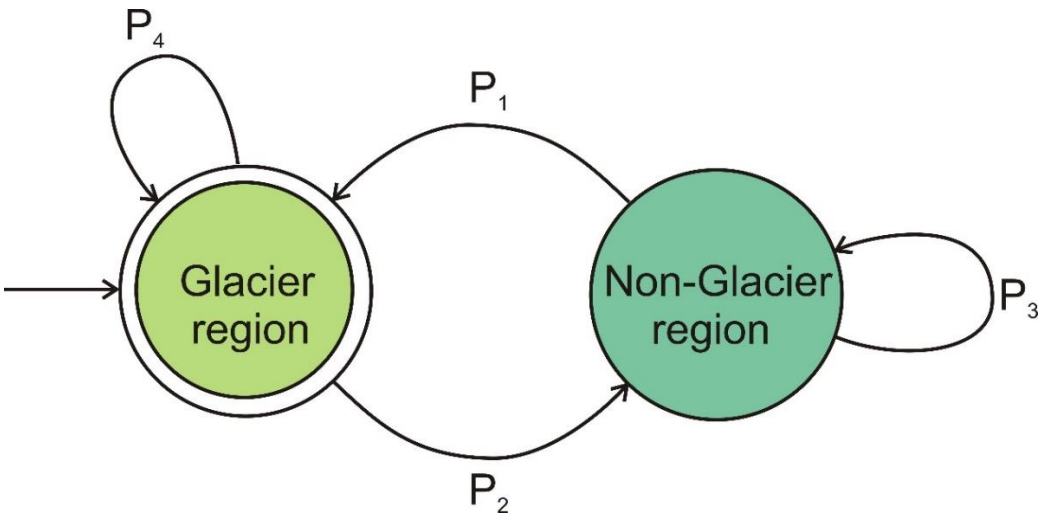

Figure 3.6 State diagram protruding all the possible transitions for this study.

The chain model says that the behaviours of the model could form a state space, that is, a list of all the possible states. For this study, the possible states are whether there is a glacier retreat phase or not. In addition, the Markov chain model tells the probability of transitioning from one state to any other state. With two states in the state space required in the study, there are a total of four possible outcomes/transitions, which are the possibility of transitioning to another state or remaining in the same state. The state diagram (Figure 3.6) cannot provide a clear understanding of modelling in practice. As a result, all of the transition probabilities are

calculated using a transition matrix. Every phase in the state space is represented by a row and a column, with each cell in the matrix indicating the likelihood of transitioning from its row to its column. To simulate the Markov chain for this study, one must follow the patterns for the past dataset (2000 to 2021 Landsat data) combined with an Artificial Neural Network to get the expected results (Zeshan et al., 2021).

In this study, ANN-CA employs a training simulation of glacier changes from 2020 to 2021 to estimate future glacier characteristics in 2022. This training scenario creates a classification model to predict a class for the new dataset (Tharwat, 2021). While the training simulation generates the glacier cover in 2021, this output will be verified using the confusion matrix (Accuracyand Kappa coefficient) by the actual glacier cover in 2022 (validation set). The training model's performance level will be determined as a result of this approach. The framework and information from the training model will be used to construct the test model once the training model reaches a high level of accuracy.

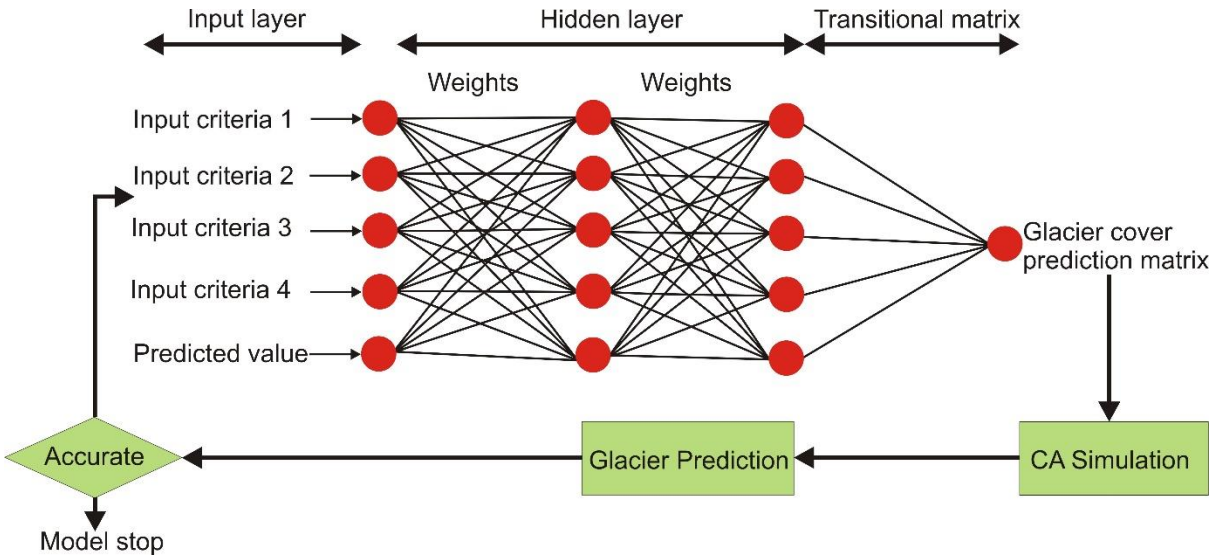

Figure 3.7 Illustration of the Artificial Neural Network-Cellular Automata paradigm.

3.4.1 Benchmark

The validation stage compares the predicted glacier area generated by the model to the reference image (the actual glacier area at the moment) and sees how the model performs. The model is validated using a range of performance indicators. The projected area that is precisely

the same as the actual data is represented by the correctness percentage, whereas the Kappa value reflects the degree of accuracy and dependability in a statistical classification (Saputra & Lee, 2019).

$$Kappa = \frac{p_o - p_e}{1 - p_e}$$

Where p_o is the proportion of observed accords and p_e is the proportion of expected accords.

The confusion matrix is another performance indicator that is used to assess the model's validity. The result if the model accurately predicts the positive class is referred to as true positive. False positive refers to when the model forecasts the positive class inaccurately. When the model predicts a negative class inaccurately, it is called False Negative. True Negative, on the other hand, occurs when the model accurately predicts the negative class. These measures can help to reduce the variance between target labels (glacier and non-glacier).

Precision is defined as the proportion of correctly classified positive samples to the total number of positive predicted samples.

$$Precision = \frac{True\ Positive}{True\ Positive + False\ Positive}$$

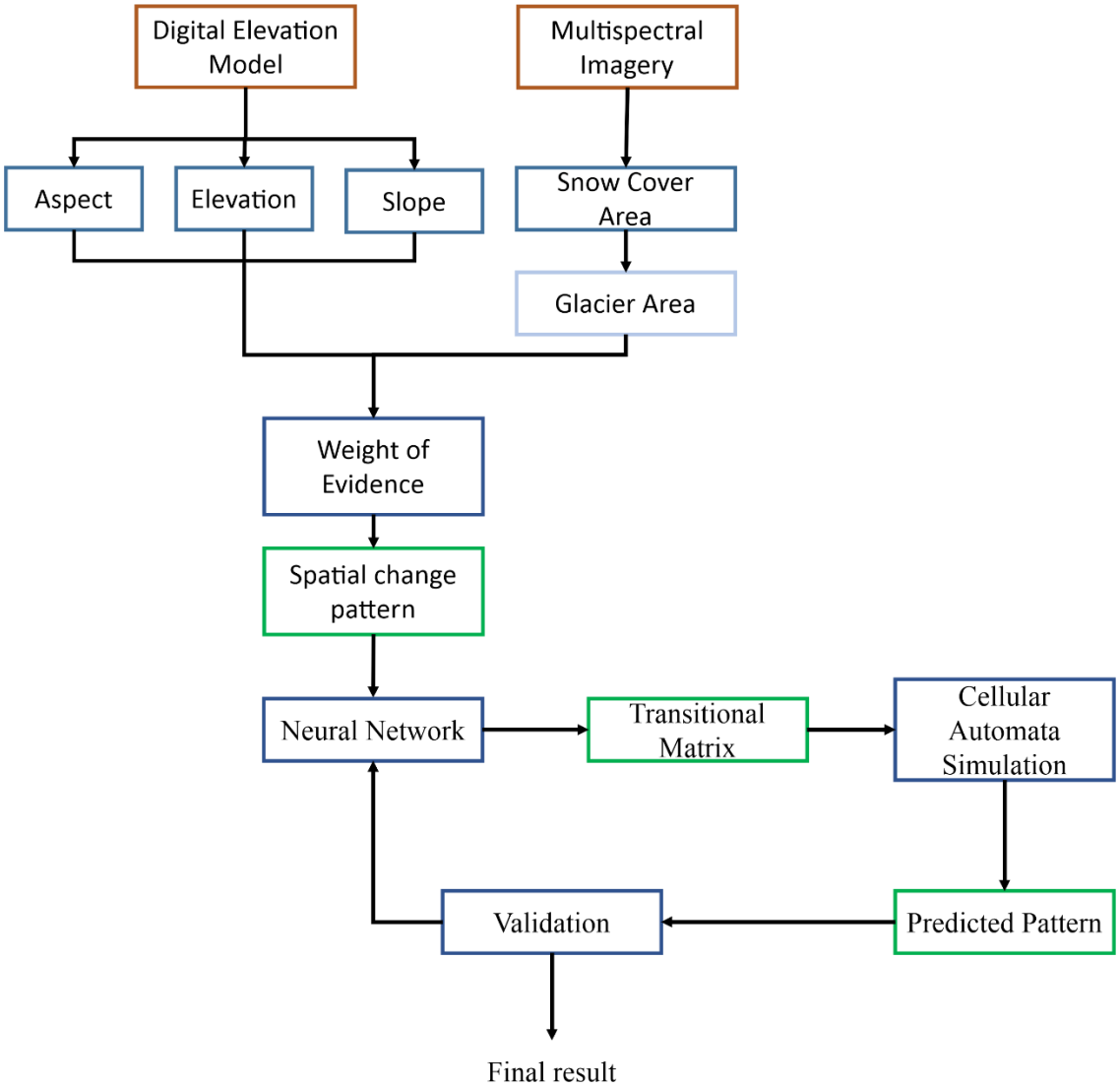

Figure 3.8 Flowchart for glacier retreat prediction.

Chapter 4. Results and Discussion

4.1 Spatial Glacier Retreat Patterns

The behaviour of glaciers has long been a hot topic in glaciology (Hock, 2005). The C value is used to indicate the outcome of the WoE method, which is used to examine the spatial pattern of glacier cover change (contrast weight of evidence). This value is comparable, allowing researchers to compare the various reactions of each spatial criterion to glacier retreat. This spatial analysis illustrates the pattern of glacier retreat over the last 9 years in this study.

4.1.1 The correlation between glacier retreat and aspect

Previous research has elucidated the impact of many factors on glacier retreats. The energy–balance concept is related to the aspect-glacier retreat relationship (Hock, 2005). Precipitation, solar radiation, and air temperature all affect glacier mass balance (Jiskoot et al., 2009). Through a relative angle to the solar zenith, the aspect can control the energy balance in this scenario, allowing the area that is highly exposed to sunlight to receive more solar radiation, resulting in the ablation process (Hock, 2005).

Glacier retreat and aspect have a C value ranging from -0.35 to 0.21. The data in Figure 4.1 illustrates the specific spatial pattern of previous glacier retreats. According to the analysis, glacier retreats have primarily occurred in the southwest and west. In all eight phases, the C values in this area are always positive. The average C values in the southwest and west are 0.17 and 0.11, respectively, which are much higher than the overall. The glacial retreat in the north, northeast, east, and west, on the other hand, was not significant. Within these zones, the C value of glacier retreat is always negative. The inconsistent response of glacier retreat in the southeast and southwest is another intriguing pattern revealed by this graph. This pattern describes the area's variable response to the glacier retreat in the Rishi Ganga basin, which can be both positive and negative. According to Jiskoot et al. (2009), this pattern is a standard procedure.

Furthermore, this figure depicts the trend response of glacier retreat, with glacier retreat in the south area showing a significant increase and a modest increase in the northwest region over the last 9 years. The increase in C value in both locations is indicative of this tendency. The diminishing effect of the C value in the northeast, east, and southeast may also imply a growing tendency of glacial retreat in this region. The change in glacier retreats on Rishi Ganga in response to aspect is related to variations in net-radiation (radiative fluxes) where the south and northwest are not protected from incoming solar radiation (Klein & Kincaid, 2006). Because of their proximity to the solar zenith, these places receive greater solar radiation. Glacier mass balance may be reduced in places facing away from the sun zenith (Jiskoot et al., 2009). Another factor is the distribution of wind-drifted snow (Jiskoot et al., 2009), which may contribute to the ablation of ice at this peak. The rapidly shrinking from the west and southwest, which may be attributable to wind distribution, is plainly visible in Figure 4.1's changing pattern.

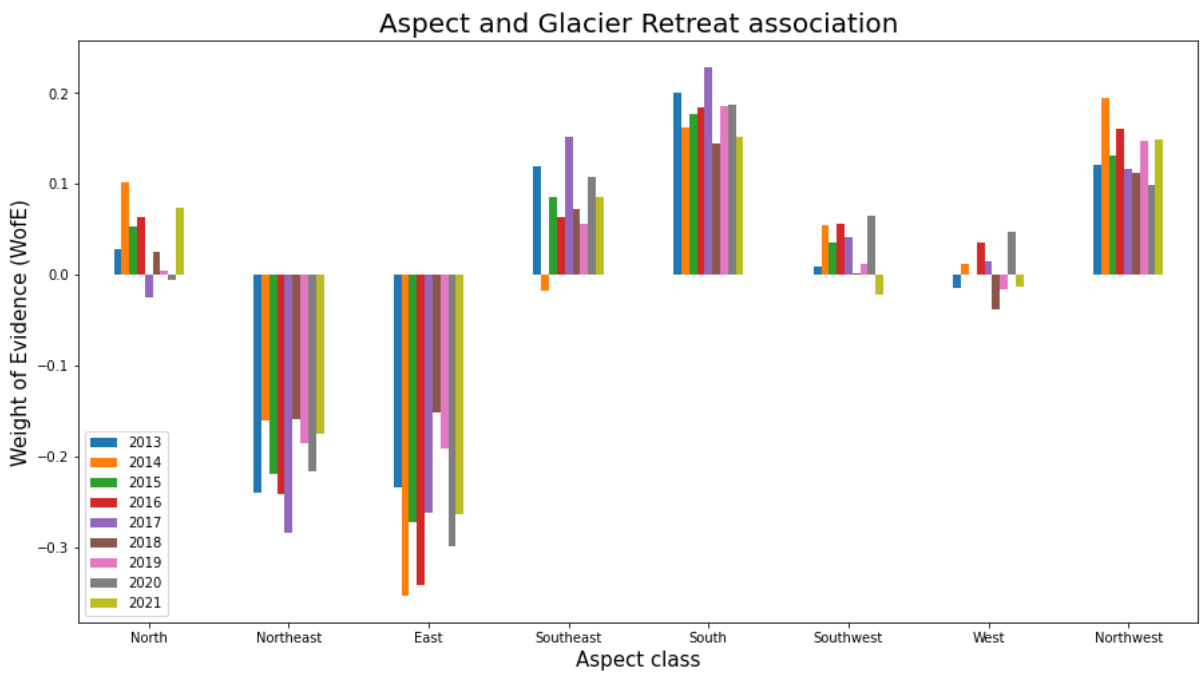

Figure 4.1 The glacier retreat and aspect relationship.

4.1.2 The correlation between glacier retreat and Elevation

The explanatory variable glacier retreat from elevation has a distinct pattern, with a propensity to retreat from lower to higher elevations (Figure 4.2). Higher elevation locations

have a higher C value (ranging from 2.3 - 2.5), whereas lower elevation places have a C value ranging from -10.3 to 0.5. The trend-line also reveals a positive tendency for the C value to increase as the region's elevation rises. This is in conformity with the results of Thompson et al. (2011), who discovered that glaciers retreat from lower to higher elevations.

The pattern of C values on the elevation criteria is understandable in general since the glacier is shrinking as it approaches the summit of the mountain. Allison & Kruss (1977) and Klein & Kincaid (1977) have both provided excellent explanations for this connection (2006). The lower elevation, when placed below the equilibrium line altitude (ELA), might begin the ablation process, causing precipitation to fall as rain. Allison & Kruss (1977) and Prentice et al. (2005) both noticed an increase in ELA elevation.

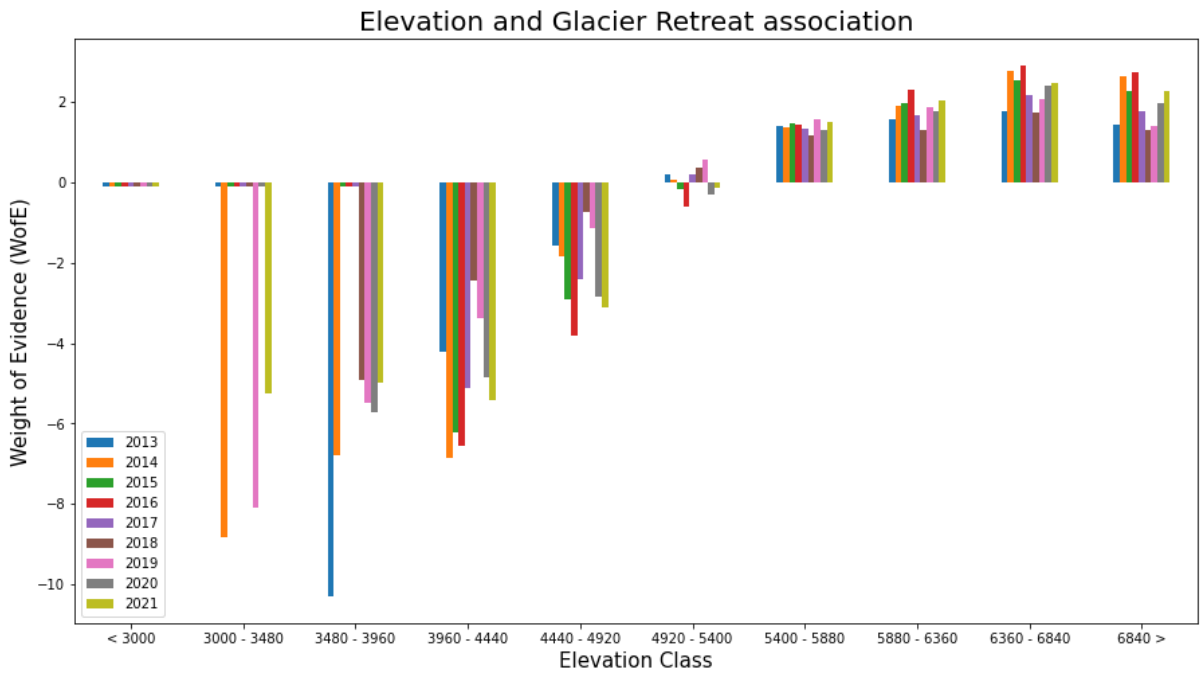

Figure 4.2 The glacier retreat and elevation relationship.

4.1.3 The correlation between glacier retreat and Slope

The influence of the slope parameter on the energy flux or energy-balance model is comparable to the association between glacier retreat and Slope (Hock, 2005; Jiskoot et al., 2009). The effect of slope on glacier retreat appears to be evident. Glacier retreat is strongly linked to slope degrees of 2 - 9 and 10 - 30, which are reflected by a high positive C value

(Figure 4.3). This slope class's average C value is 0.46 and 0.41, respectively. This suggests that the glacier retreat on Rishi Ganga Basin is controlled by the moderate sloping slope (2-9) and steep sloping slope (10-30). These figures outperform the overall average of 0.15.

The historical pattern shows an increasing trend of glacier retreat occurring in an area with a slope degree of 10 - 30, while the trend of glacier retreat in a slope degree of 30 - 40 is decreasing. The C value for the 2018 period in the 2 – 30 degree slope demonstrates a strong positive relationship (C = 0.63). The C value in other slope classes, on the other hand, is extremely negative, as shown in Figure 4.3. This means that glacier retreat is distinctive in areas with level plains (0 - 20), mildly sloped (3 - 40), and very steep slopes (>40).

Figure 4.3 The glacier retreat and Slope relationship.

The influence of this criterion can be seen in the variable response of slope on glacier retreat. The main contributor to glacier retreat is radiative flux, which is heavily influenced by slope and aspect. The influence of slope on temperature stratification, which leads to glacier melting, is mentioned by Hock (2005). Through albedo reflection, the slope is also responsible for radiative fluxes. This study indicates a positive relationship between glacier retreat and flat area, which is consistent with the earlier research that showed glacier retreats occurred primarily in a lengthy flat area (slope 0 - 20) and declined as the slope became steeper (Jiskoot,

2009). Because of its proximity to the solar zenith, the flat area receives more solar energy. The irregularity in this pattern reveals a location component that is crucial in the Rishi Ganga basin but has yet to be discovered.

4.1.4 Glacier History and Future Prediction

Looking at the glacier history of the Rishi Ganga region, the past data shows a variable shift in glacier patterns. Even though there was variable shift over time, this analysis demonstrates the substantial loss of snow cover. Because the images were not all acquired at the same time, the changes could be attributable to seasonal factors. The figure 4.4 shows a decrease in glacier area from 18665.56 Ha to 10479.88 Ha from 2013 to 2021, and looking over the best fit even without the forecasting glacier area for upcoming years can be said to show a decreasing trend. The recessional trajectory is also diminishing with time, according to this investigation.

The rate of shrinking was not uniform across the peaks, and other glaciers exhibit the same variation in local glacier retreats. The phenomenon could be produced by a geomorphological variance state, resulting in glacial retreat variability. Nanda Devi and Trisul glaciers account for the majority of the glacier area in the Rishi Ganga region. In the last nine years, the glacier on Trisul summit has accounted for 15.44% of the total glacier area on the Rishi Ganga region, where as the glaciers in Nanda Devi and Dunagiri take about 8.82% and 6.6% of the overall glacier region. The substantial downward trend on Nanda Ghunti is observed to continue till 2021. Another conclusion of this study is the large-scale disappearance of the Dunagiri glacier over a nine-year period. The abrupt surge in glacial area was noted near the year 2017-18, but it has since begun to decline over time.

Due to the wide range of colour changes in the area, there was a considerable decline in glacier areas from 2016-17, especially on Western Dunagiri and Northern Nanda Devi, as shown in Figure 4.6. This finding corresponds to the data in Figure 4.5, which indicates a significant decline in both of these peaks in 2017. The spatiotemporal shift is proceeding in an East-West orientation. The majority of the glaciers that were receding were in the Eastern

corner of the entire glacier system. Figure 4.6 clearly shows the significant loss in number of glaciers in the past in Dunagiri and Trisul region.

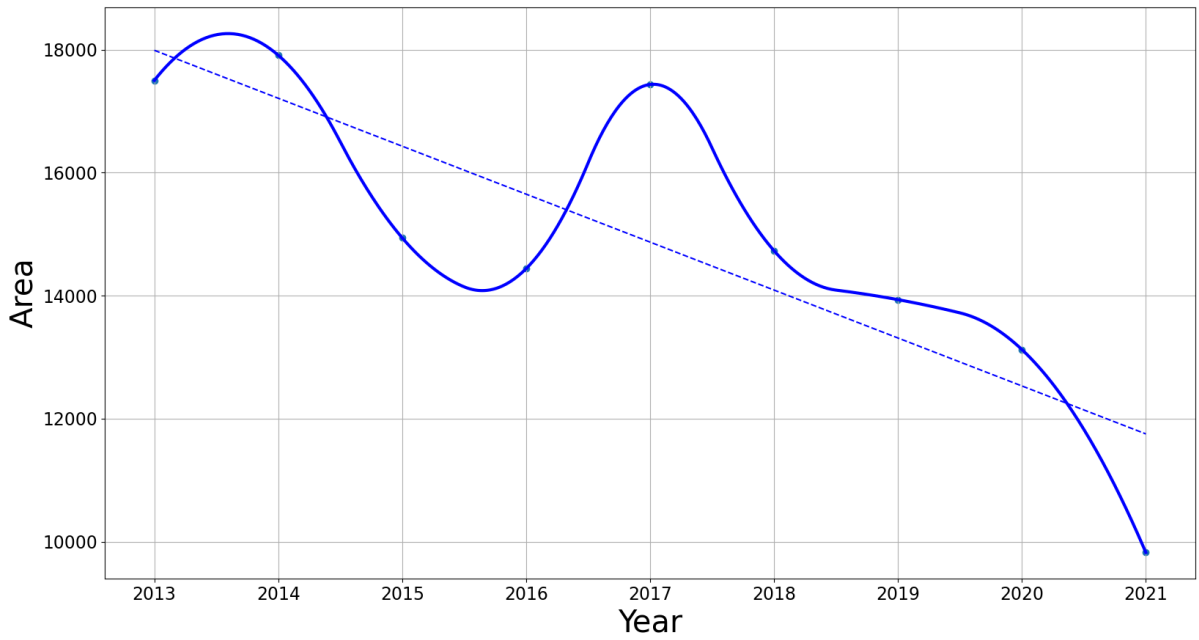

Figure 4.4 The historical glacier cover on Rishi Ganga Basin from 2013-2021. The linear regression value represents the changing trend of the glacier region.

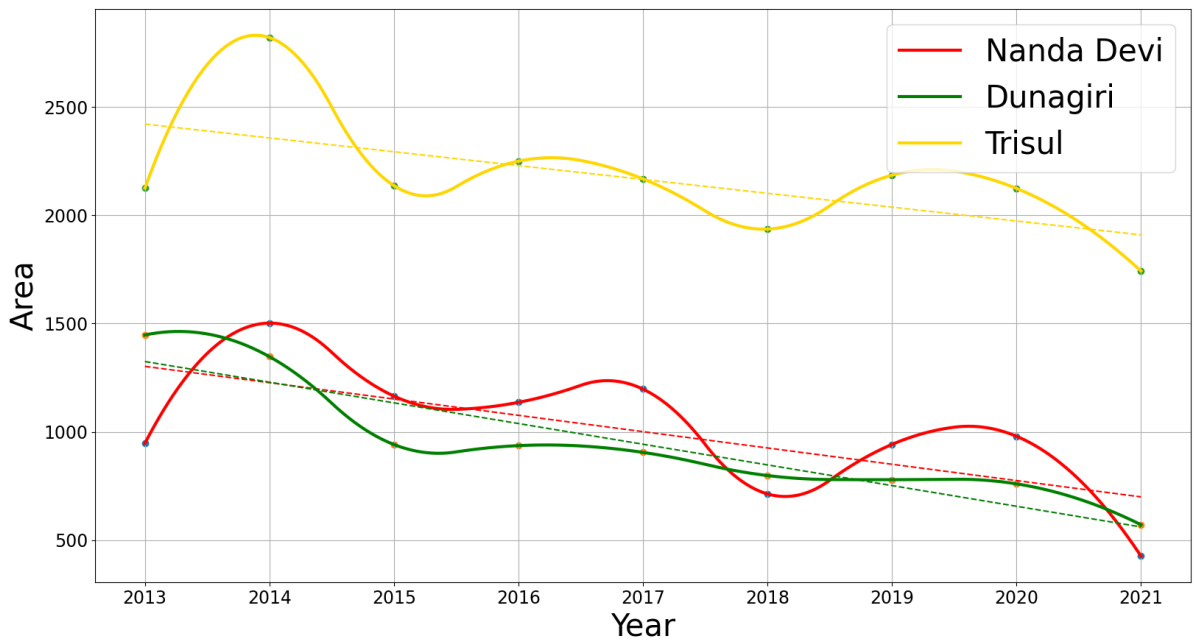

Figure 4.5 The history of glacier changes on each summit depicted from 2013 to 2021. The linear regression value represents the changing trend of the glacier on each summit.

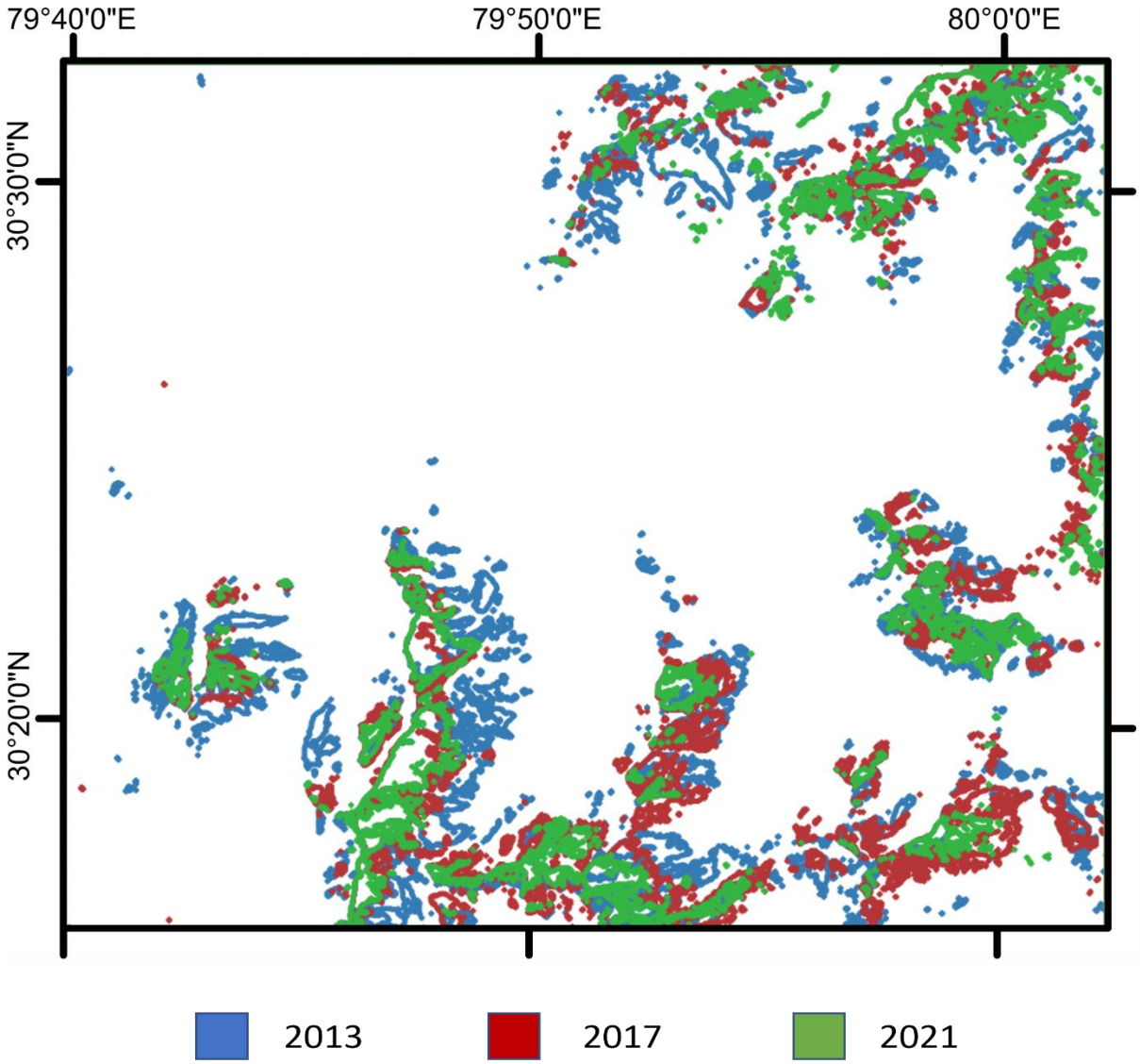

Figure 4.6 The spatiotemporal change of glacier cover over the Rishi Ganga Region over the time period of 2013, 2017 and 2021. The gradation colour change from blue to red to green might be viewed as a changing trend.

The study continued on to predict the future glacier retreat using a neural network. The predictions initiate with the training of the data model using historical glacier maps (before 2021) and morphological feature dataset while trying to predict the 2021 dataset. Training helps increase the likelihood of predictions; more the accuracy, better the transition matrix obtained as the end product of the neural network model. This transition matrix is then used in cellular automata methodology to create the learning rules to predict the classification, glacier and non-glacier section of the matrix. The outcomes of the model transition analysis by ANN indicated that the three dependent factors, aspect, elevation, and slope, have a considerable influence on changes in glacier cover. The ANN model was modified using multiple hyperparameters for

this study, and the results indicated that a five-layer model was able to provide relatively high accuracy for the transition matrix, with an accuracy of up to 89 percent. For the current data, the degree of consistency (Kappa coefficient) was calculated to be approximately 0.87.

Table 5. Model confusion matrix.

		Predicted Value		
Observed Value		Glacier (1)	Non-Glacier (0)	Total
	Glacier (1)	82562	141873	224435
	Non-Glacier (0)	12336	949456	961792
	Total	94898	1091329	1186227

The ANN-CA model predicts a consistent decline pattern for future glacier cover. This study agrees with the IPCC's findings, which state that glaciers will continue to melt for decades or centuries (IPCC, 2021). According to this analysis, the glacier on Dunagiri would melt entirely by 2027, while a substantial portion of the glacier on the eastern side of Trisul and Nanda Devi may melt around 2030. This forecast aligns with an ice core data-based projection that claimed the massive ice loss in 2026. This study can reconstruct the future spatial changes in glacier cover in the Rishi Ganga Basin.

The ANN-outstanding CA's performance in this simulation demonstrates that it can be particularly effective in modelling future changes in glacier retreats. This research indicates that ANN-CA may be utilised for a variety of environmental issues, including land use/land cover and urbanisation. Unlike early studies, this updated ANN-CA model takes into account

overfitting and linear fluctuations, and it has a high precision. The ANN-CA is unable to identify changes in seasonality patterns, which is a drawback in the existing model.

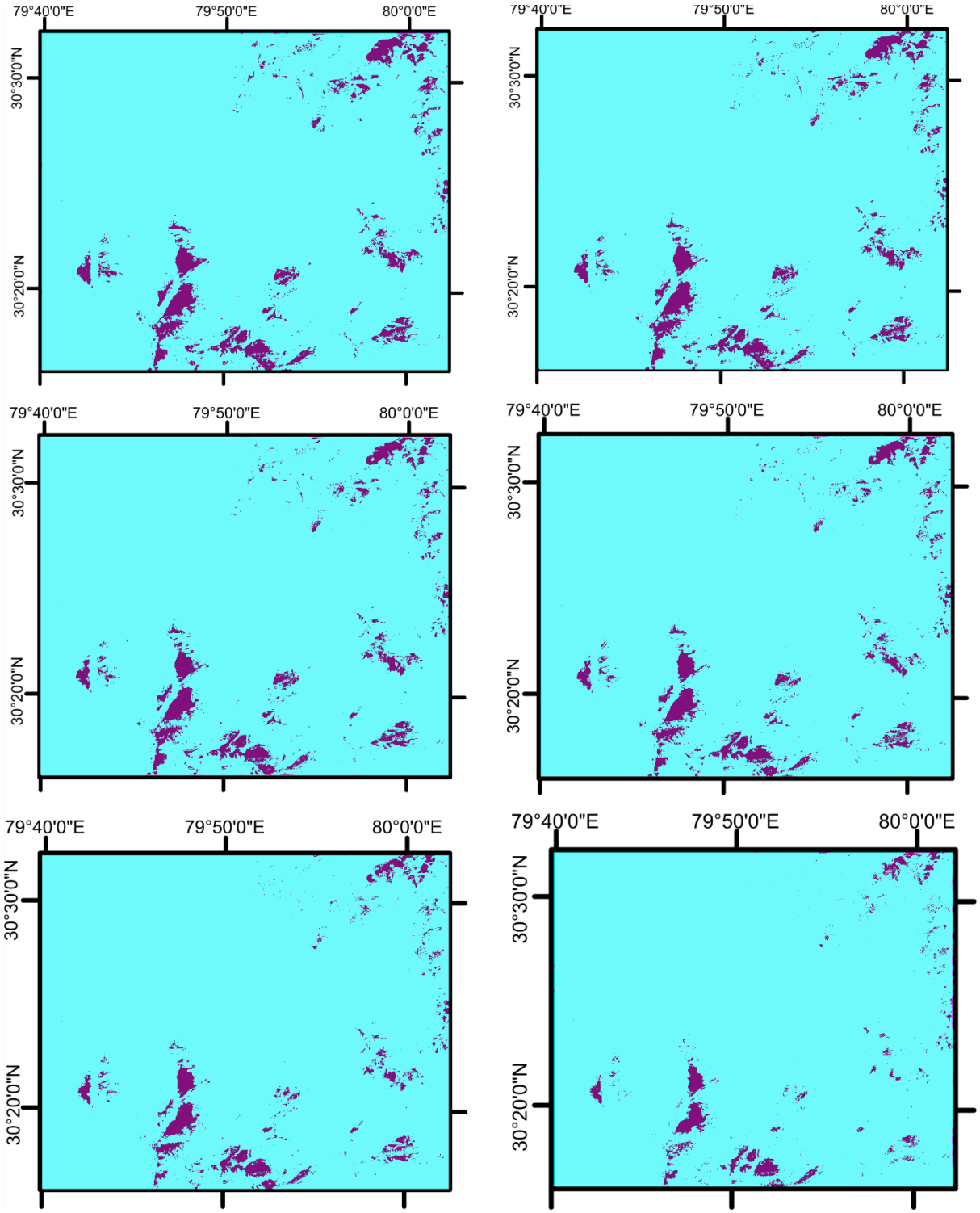

Figure 4.7 Future glacier predicted using model in Rishi Ganga Basin.

Chapter 5. Tools Developed

A QGIS plugin called "Glacier Delineation Simulation" was used throughout this research. This was an attempt to assist researchers in their studies, namely in the areas of snow-cover variability evaluation and snow-melt runoff modelling. The plugin is also free and open-source, and it can be found at https://github.com/Sar1hak/Glacier_retreat. It includes algorithms for manipulating and analysing DEM data, calculating weight of evidence, and applying modelling techniques to yearly Landsat datasets. The plugin was written entirely in Python, and the tools are only available in Python.

Apart from the QGIS plugin, all of the supplementary scripts and Jupyter Notebooks developed during this study are accessible to everyone at https://github.com/Sar1hak/Glacier_retreat.

5.1.1 Development

Following these steps will allow the user to access the plugin:

1. Using a command-line git interface, running the following command will download the repository:
   git clone https://github.com/Sar1hak/Glacier_retreat.git
2. Once downloaded, the plugin can be directly accessed via the run.exe file. Or the user can open the terminal and run the GDS_GUI.py (Note: Do see the requirements.txt file for the appropriate execution of algorithms).

Based on the intended datasets to be processed, the tasks performed by the plugin are classified into four categories. The tasks concern the processing of the Digital Elevation Model and Snow Cover, the degree of spatial association, and glacier retreat modelling. DEM classification, Glacier processing, the weight of evidence, and Model simulation are the tabs for these processing tasks, respectively.

5.1.2 DEM classification

The DEM classification tab deals with DEM attribute user-defined The DEM classification tab is for user-defined DEM attribute classification. The study-specific DEM file can be uploaded by selecting the 'Upload' button, which displays a dialogue window where individuals can navigate to the file's directory.

The second section in the tab deals with the Attribute selection, which can be selected from the dropdown menu. If a user already has the attributes in raster format, the application presents an alternative method of uploading information, which requires the checkbox to be chosen beforehand. A dialogue box may appear during the variable selection procedure that must be filled in to set the variable range during the classification process. In the table presented in the tab, one could see the selected attributes. The application currently only interacts with three variables: Aspect, Elevation, and Slope, however, additional variables may be included in future releases.

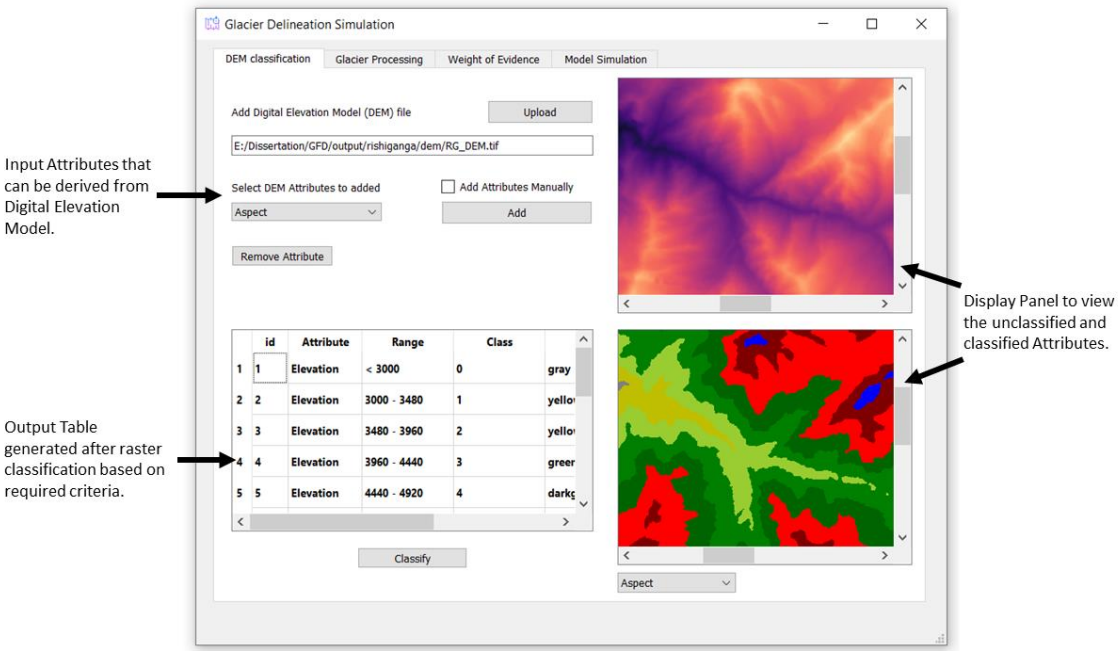

Figure 5.1 DEM classification tab, and Elevation classification output

The classification process commences with a click on the 'Classify' button, which in turn, after assessment, shows the original and classified variables in the Display panel for the tab. The display panel displays feature one at a time, and the dropdown menu below the display panel

allows users to access additional attributes. During the processing, the table is updated to include all the range values, new classes, and the colour each category represents in the display panel.

5.1.3 Glacier Processing

During this study, a cloud-free glacier cover dataset was generated based on Dorothy K. Hall et al. 2002 NDSI algorithm. Currently, this tool allows users to obtain glacier cover using the same algorithm that was used in this study. The algorithm requires a Multispectral image dataset as input, and the user can provide a complete folder for the processing or an individual file. The collected files will be displayed in the output table for the Glacier processing tab.

The output table shows all the files taken into consideration for the processing and the year of the associated data file. The processing can be started by clicking on the 'Calculate' button, which begins the correction algorithm for the files. If any is required, geometric correction, atmospheric correction, and radiometric correction are only considered. Afterwards, as seen in section 3.2, the corrected data is shifted to the NDSI assessment, followed by the flags set by Dorothy, 2002.

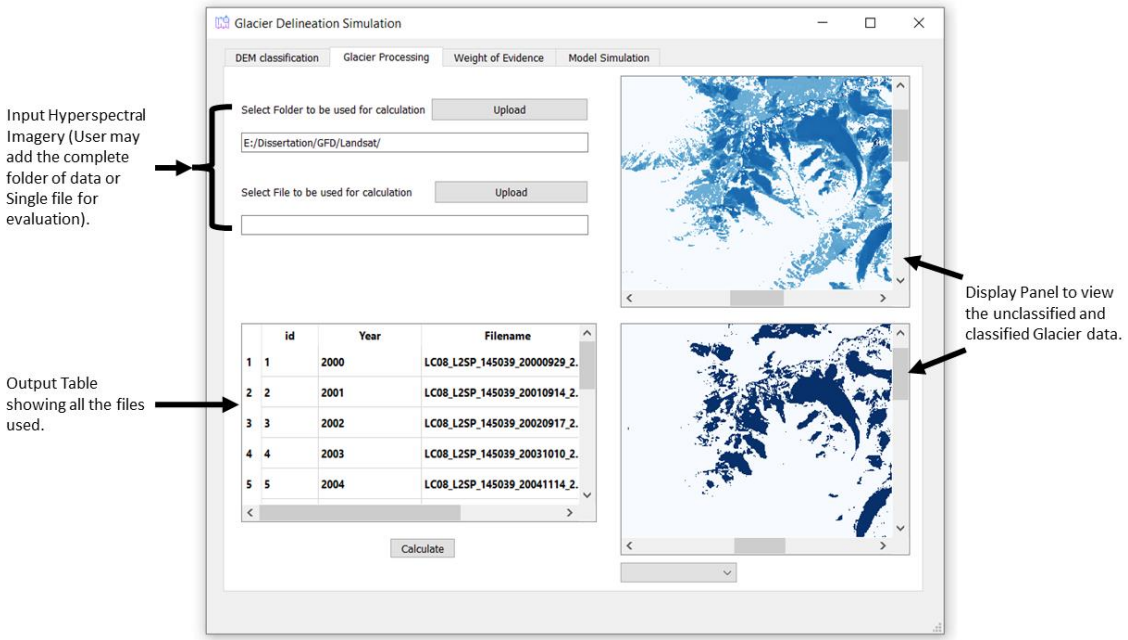

Figure 5.2 Glacier Processing tab, and NDSI classification output.

5.1.4 Weight of Evidence

A separate section was added to get an overview to get a clear correlation between the different input criteria and the glacier cover. The correlation was predicted through the occurrence and non-occurrence of the glacier event on a pixel scale. The 'Weight of Evidence' tab displays an entire table of different classes in the input criteria and the necessary variables that are required to find the association between the dataset and glacier cover. The table gives a complete statistical interpretation of all the input parameters.

The application provides a visual bar graph representation of the different classes of input parameters and their corresponding values of the weight of evidence 'C', to get an overview of the strength of each class over glacier value. Both the evaluated data table and graphs can be saved in the user-defined location by selecting the save button for the respective set.

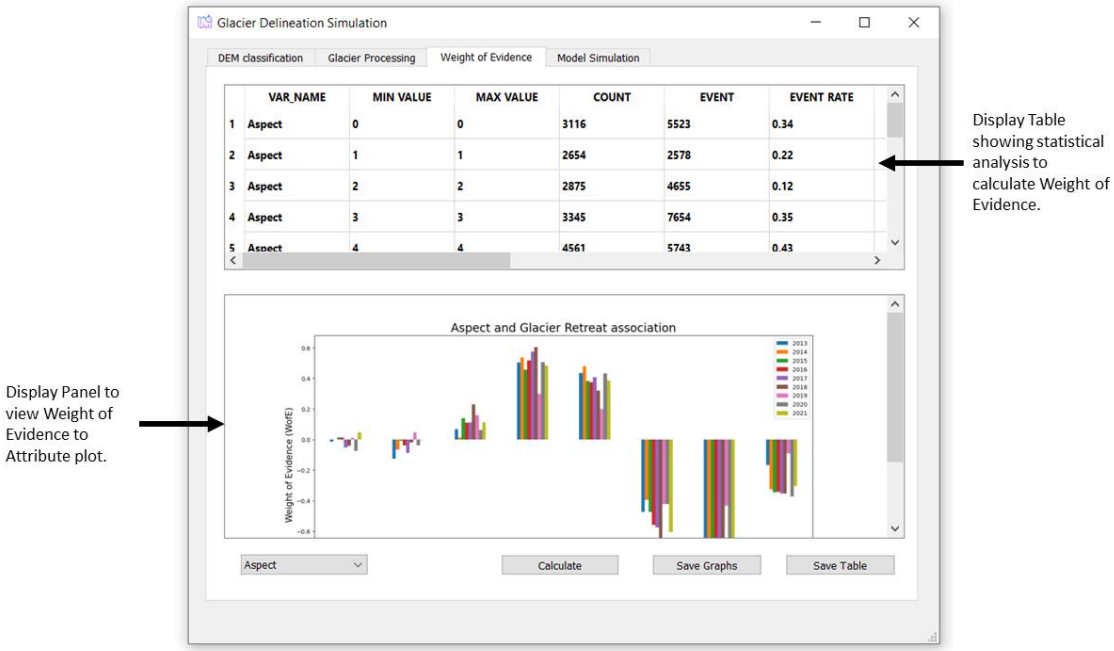

Figure 5.3 Weight of Evidence tab, and output table and plot for all the associated attributes.

5.1.5 Model Evaluation

The last tab in the application is the 'Model Evaluation' tab, which evaluates and generates a glacier image forecasted for the near future. The model currently takes the help of the Cellular Automata simulation to predict glacier retreat maps; depending on the number of iterations, the forecasted time period can be increased further.

As seen in section 3.4, the model is generated using a neural network that provides a Transition matrix that helps the Cellular Automata converge the glacier model further for better future prediction. Similar to 'Weight of Evidence', a saving option is provided to the user to generate a feature map. The Display panel shows the iterated map with the output year below it for better understanding.

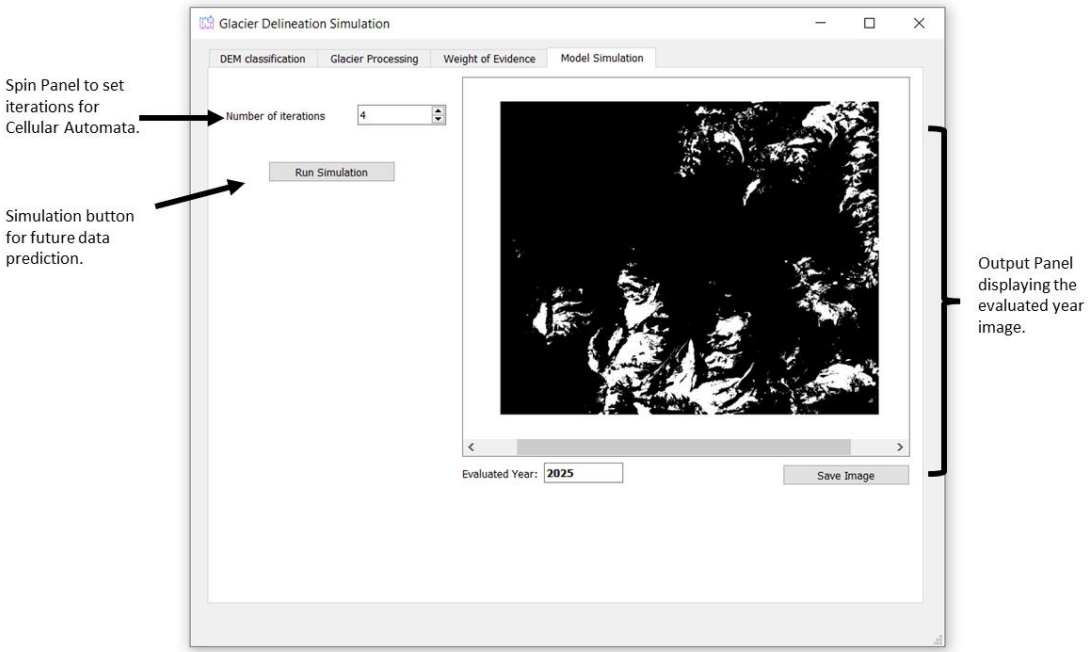

Figure 5.4 Model Simulation tab, and neural network cellular automata image evaluation.

Chapter 6. Conclusion

Using a series of Landsat images, this study was able to explain the evolution of a glacier in the Rishi Ganga Basin during the last nine years (2013-2021). The glacier had already receded by 56.1 percent of its 2013 extent by 2021. The spatiotemporal analysis was carried out to better comprehend the system's dynamic behaviour. The glacier retreat on the Rishi Ganga Basin was intimately correlated to elevation, followed by aspect and slope, among the three spatial criteria examined in this study. Through the radiative fluxes/energy-balance model and elevation-linear equilibrium, these three factors have a significant impact on the retreating process. Furthermore, this research accurately predicted future glacier cover on the Rishi Ganga Basin (degree of correctness 89.6% and Kappa 0.87). As a result, this research reveals that ANN-CA may be used to predict glacier decrease. The results show decrease in the glacier cover in Trisul and Nanda Devi, while complete melt down of glaciers in the Dunagiri mountains. While Artificial Neural Network with Cellular Automata is a great method for forecasting but insufficient data may cause overfitting of data and decrease in accuracy. The application deviced for the study can improved further in machine learning aspects for better accuracy of the outcomes.

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